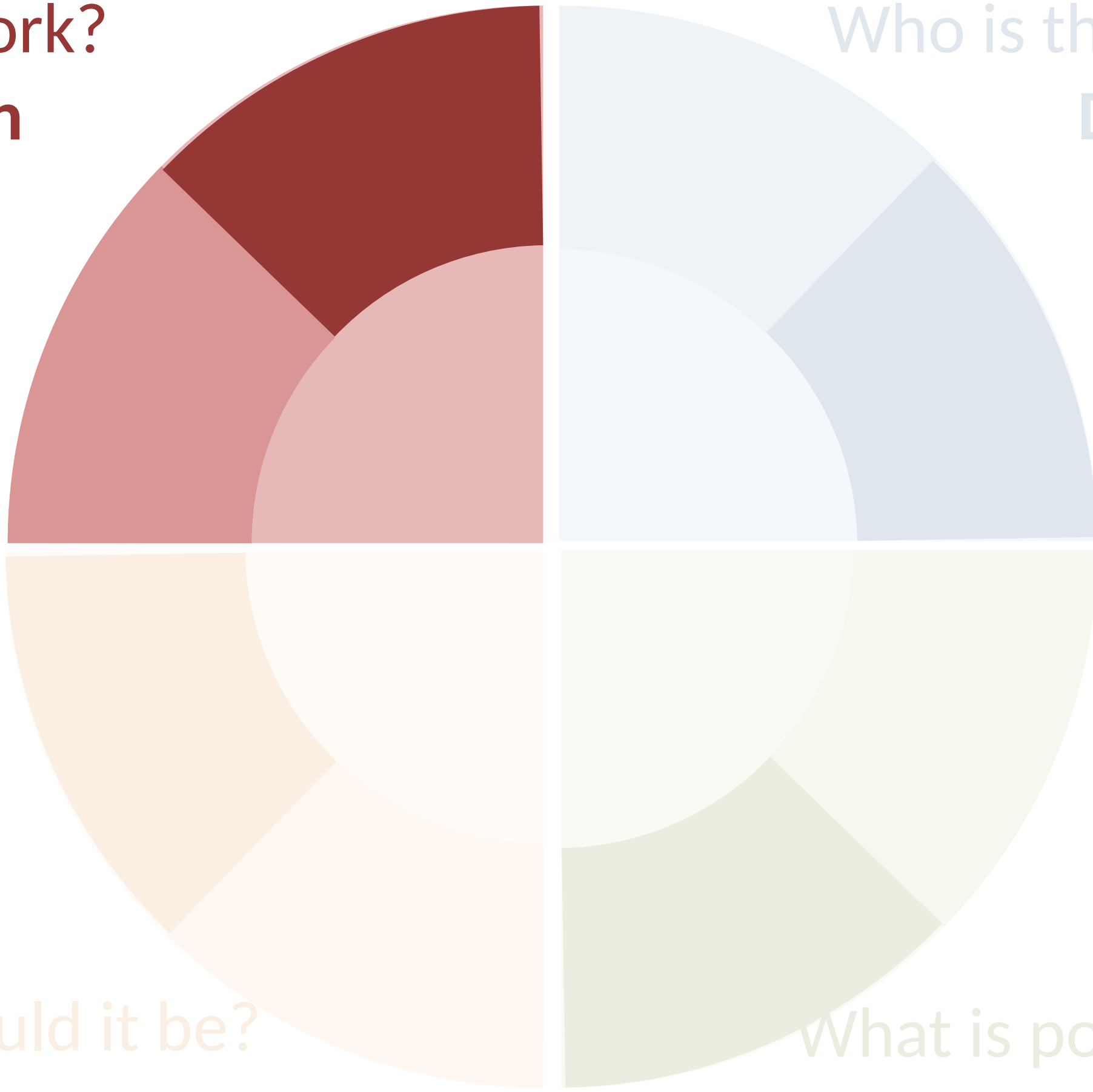


Does it work?  
**Evaluation**

Who is the user?  
**Discover**

What should it be?  
**Design**

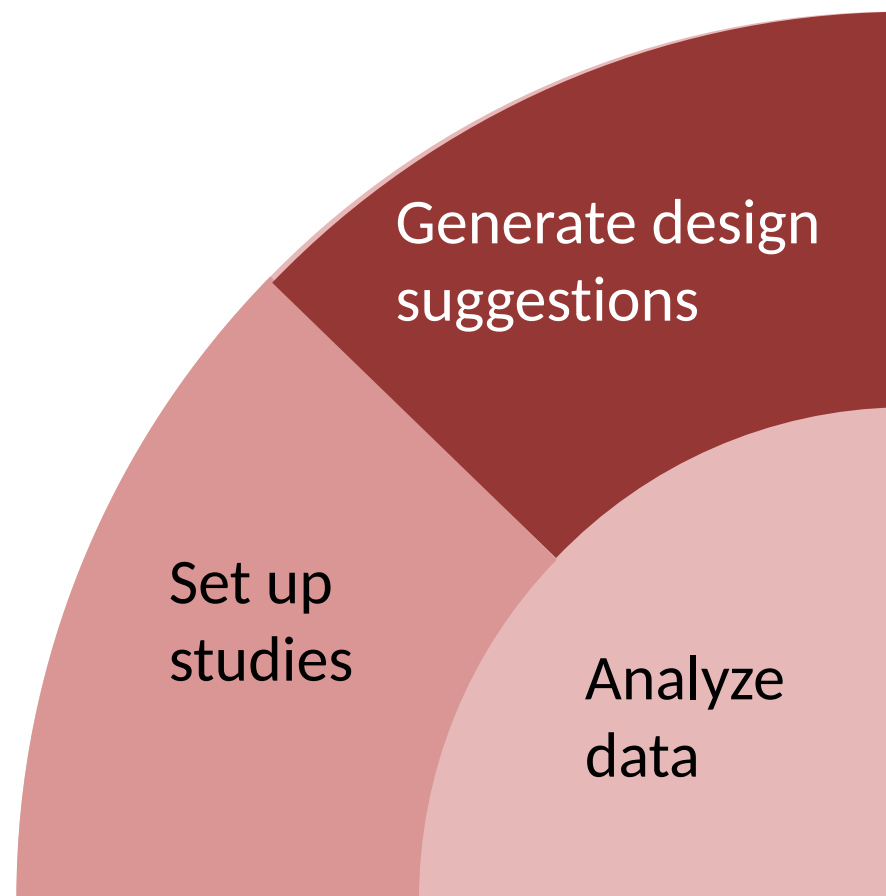
What is possible?  
**Invention**



# Does it work? Evaluation

## Design Resources

- list of problems found
- implications for re-design



## Collect information

- Empirical evaluation
- Analytical evaluation

## Analyze information

- statistic analysis

# Why evaluate?

Ensure that the system matches users' needs or usability goals.

To discover problems or judge system functionalities

- Does it facilitate users' tasks?
- Does it offer the right features, easy to reach, and presented as expected?

To judge effects on users (Understand users' behaviors)

- How easy is the system to learn and use?
- How do users feel about the system?
- Are there areas that overload users?

To understand and predict users' behaviors

# Evaluation Methods

## Various methods

- Lab studies vs. Field studies
- Qualitative vs. Quantitative studies
- With or without users

# Where to evaluate:



Laboratory



Field

# Where to evaluate: Laboratory



- + **Equipment**  
(audio / video, see-through mirrors, special computers)
- + **no disruptions**, quiet
- + Only place possible if real use dangerous, remote (ISS...), or **controlled** situation needed

Natural **environment missing**  
(shelves, wall calendar, streets, people...); unnatural situation (relevance?)

# Where to evaluate: In the field

- + Situations (location and context!)
- + Behavior more natural
- + More realistic  
(also because of disruptions)
- + Better suited to long-term studies

Noise, task interruptions

Will still feel like a test situation



# Resources

<http://www.usabilitynet.org/tools/methods.htm>

you can select the most appropriate methods depending on three conditions



limited time/resources



No direct access to users



Limited skills/expertise

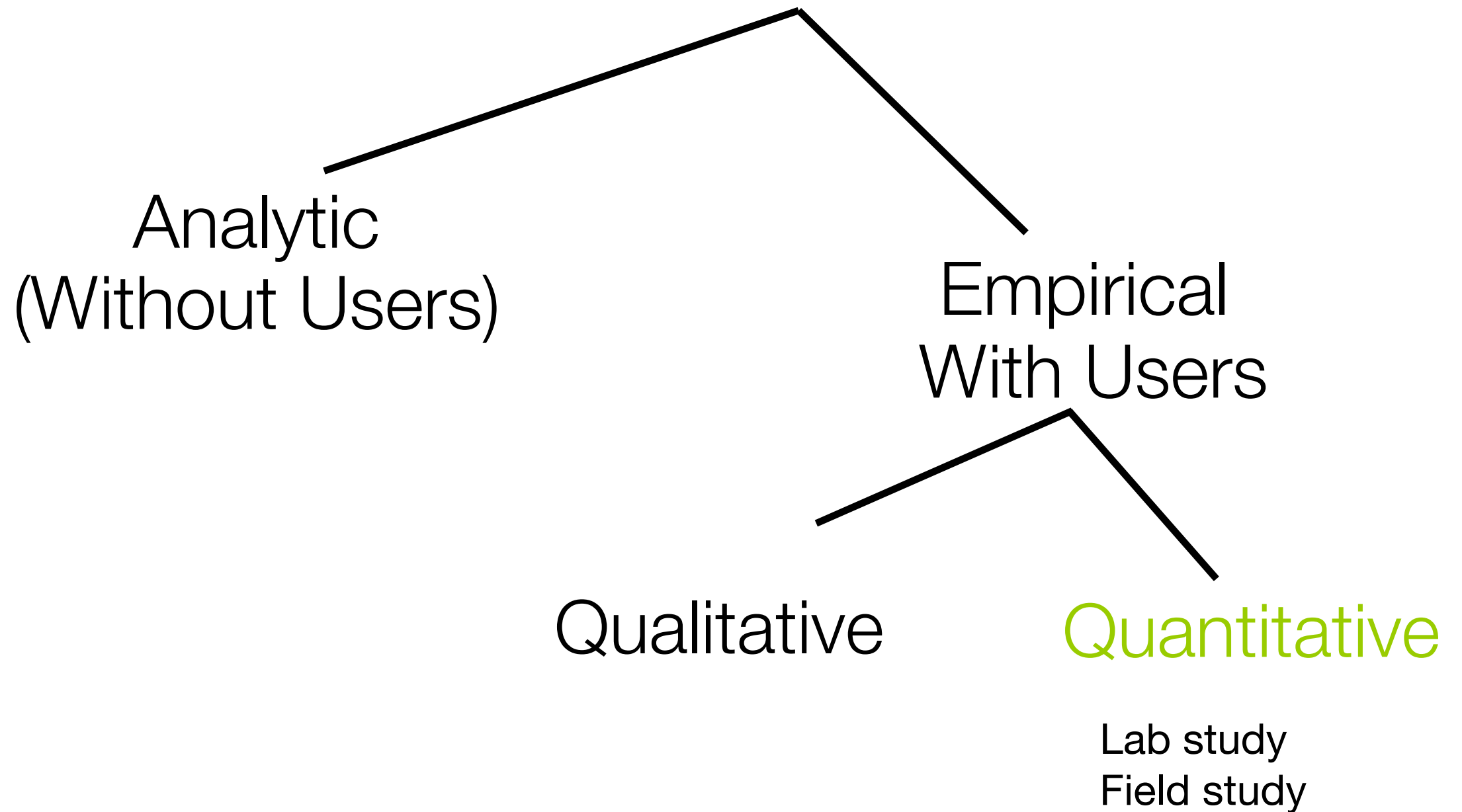
| Planning & Feasibility | Requirements                | Design                    | Implementation    | Test & Measure               | Post Release          |  |
|------------------------|-----------------------------|---------------------------|-------------------|------------------------------|-----------------------|--|
| Getting started        | User Surveys                | Design guidelines         | Style guides      | Diagnostic evaluation        | Post release testing  |  |
| Stakeholder meeting    | Interviews                  | Paper prototyping         | Rapid prototyping | Performance testing          | Subjective assessment |  |
| Analyse content        | Contextual inquiry          | Heuristic evaluation      |                   | Subjective evaluation        | User surveys          |  |
| ISO 13407              | User Observation            | Parallel design           |                   | Heuristic evaluation         | Remote evaluation     |  |
| Planning               | Context                     | Storyboarding             |                   | Critical Incidence Technique |                       |  |
| Competitor Analysis    | Focus Groups                | Evaluate prototype        |                   | Pleasure                     |                       |  |
|                        | Brainstorming               | Wizard of Oz              |                   |                              |                       |  |
|                        | Evaluating existing systems | Interface design patterns |                   |                              |                       |  |
|                        | Card Sorting                |                           |                   |                              |                       |  |
|                        | Affinity diagramming        |                           |                   |                              |                       |  |
|                        | Scenarios of use            |                           |                   |                              |                       |  |
|                        | Task Anaysis                |                           |                   |                              |                       |  |
| Requirements meeting   |                             |                           |                   |                              |                       |  |

# Evaluation Methods

## Various methods

- Lab studies vs. Field studies
- Qualitative vs. Quantitative studies
- With or without users

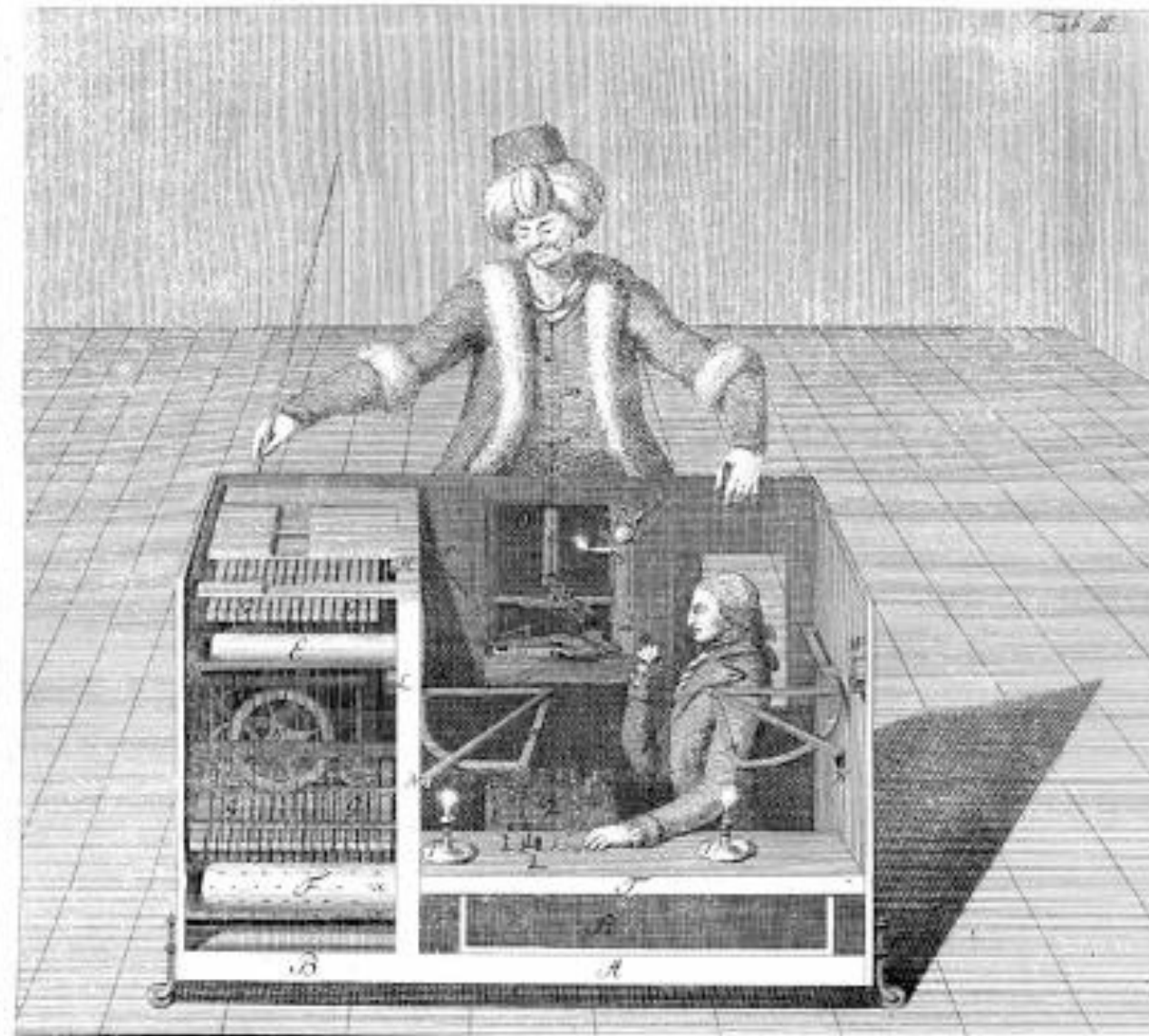
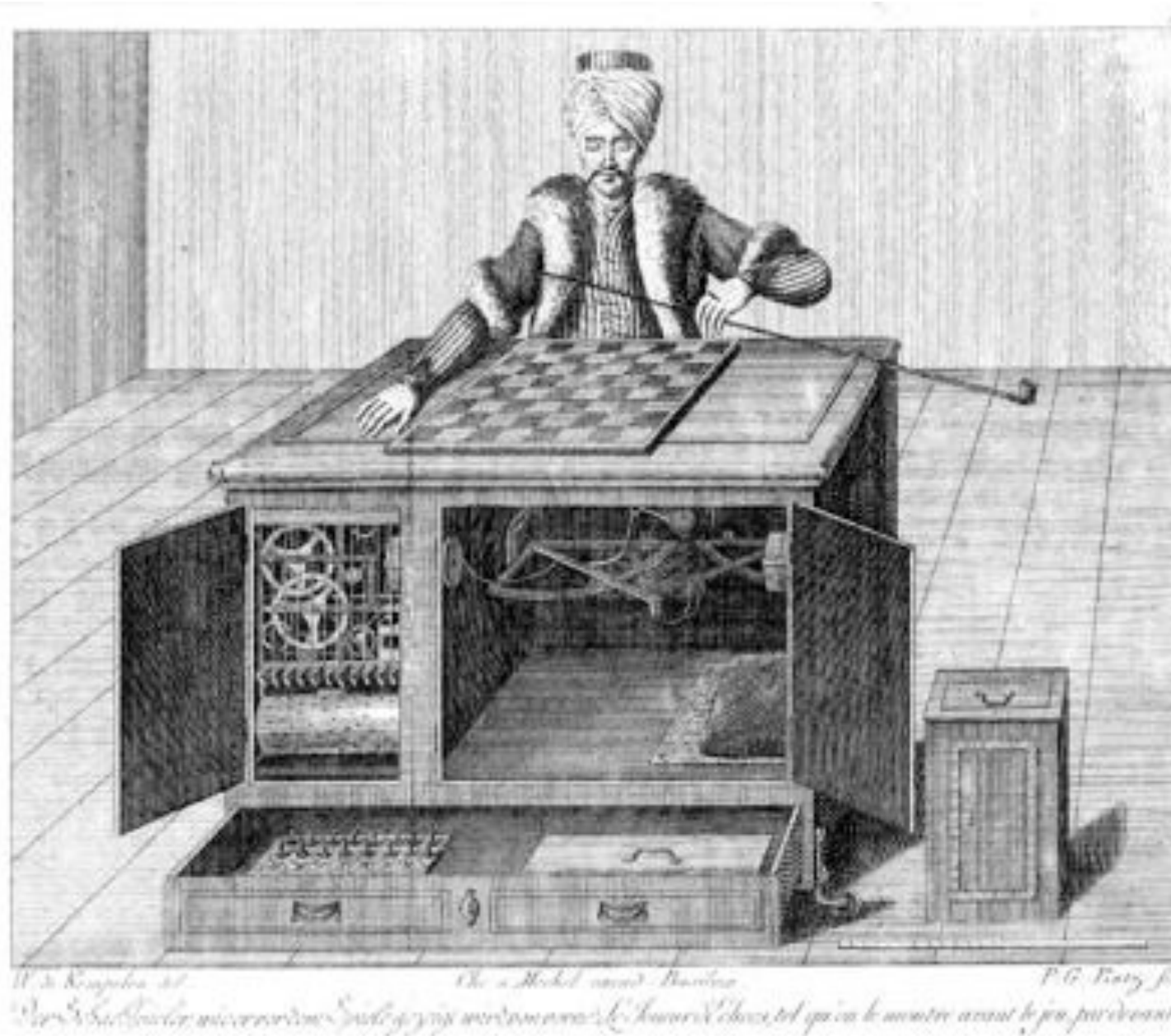
# Evaluation Methods



# Wizard of OZ

The first “Chess Computer”

– “Wolfgang von Kempelen (1769)



# Wizard of Oz

Method for testing a **non-existing** system

Human “wizard” **simulates** system responses

Complex functionalities

- speech and gesture recognition,  
language translation

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What the user sees

# Wizard of Oz

Method for testing a **non-existing** system

Human “wizard” **simulates** system responses

Complex functionalities

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language translation



What the user sees



The Wizard

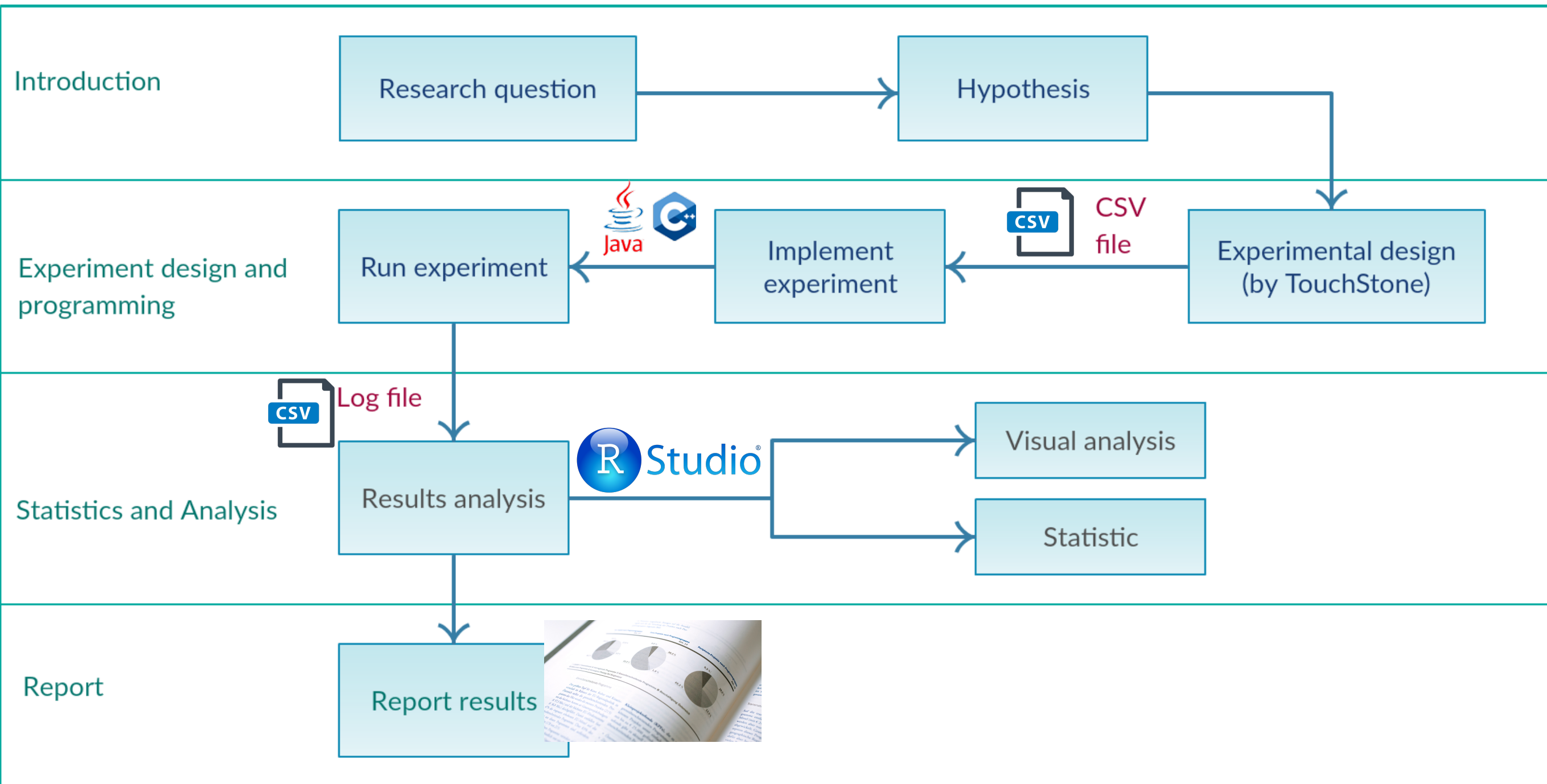
# Controlled experiment

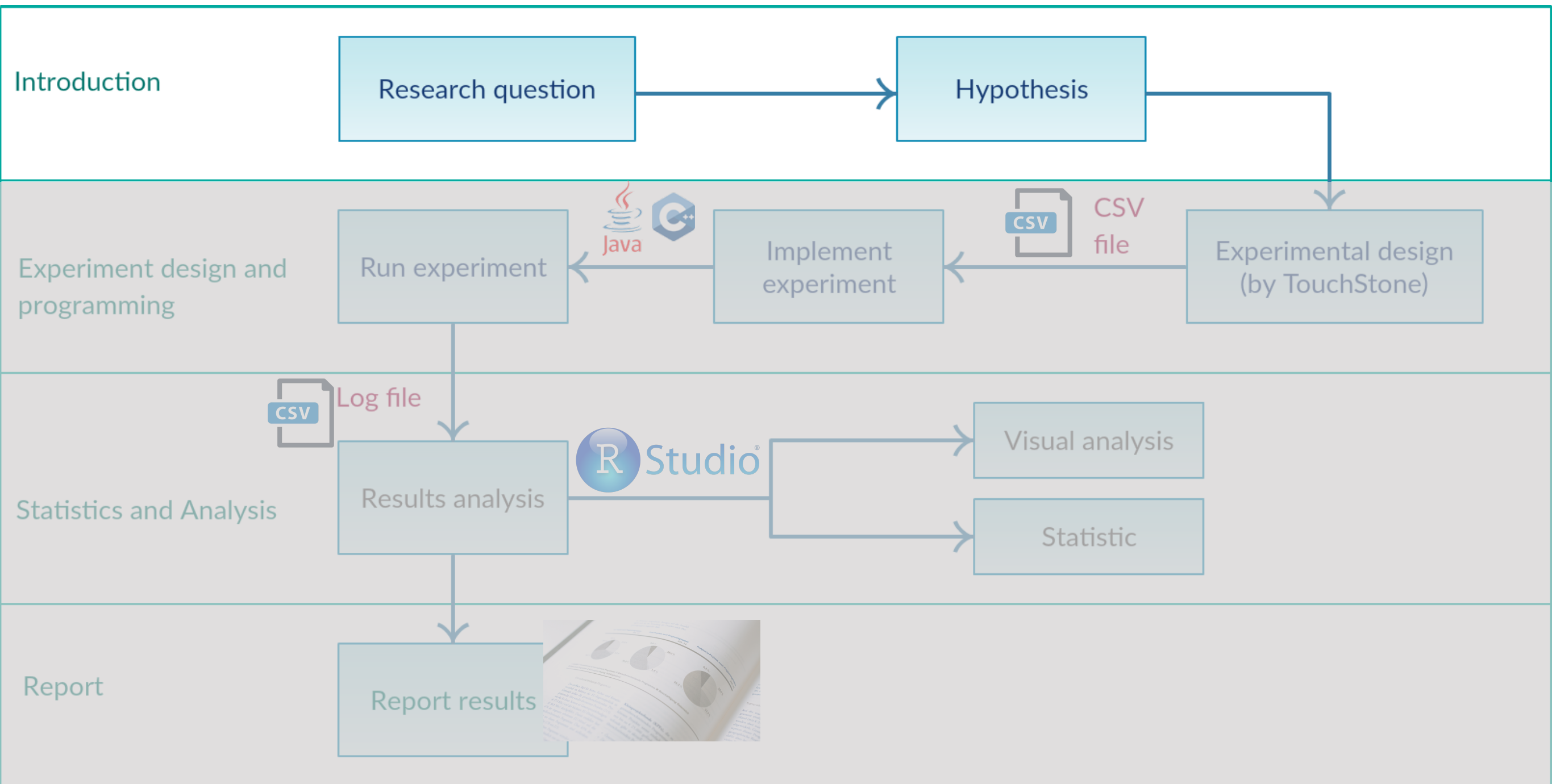
“A test under controlled conditions that is made to demonstrate a known truth, examine the validity of a hypothesis, or determine the efficacy of something previously untried”

*Mouse is faster than touchpad*



# Outline





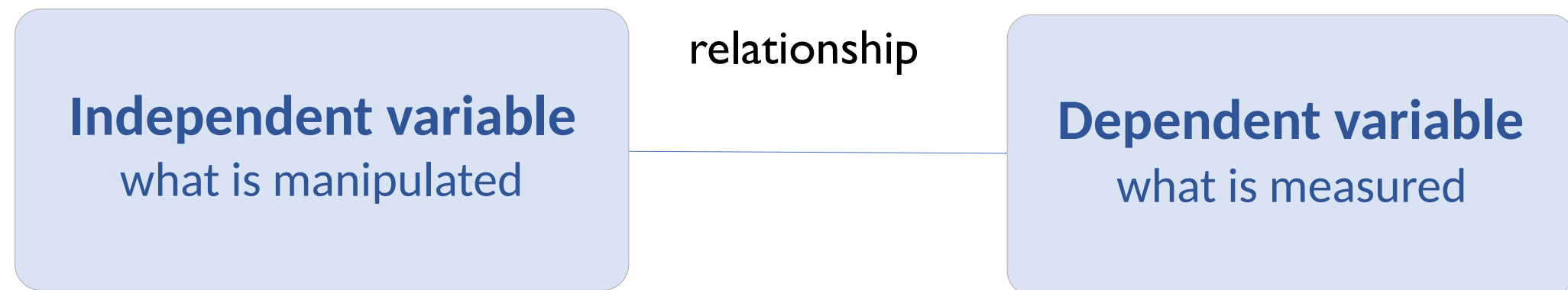
# Hypothesis

A claim that predicts outcome of experiment

- Example: Reading text in capital letters takes longer than in reading text in small letters
- Mouse is “better” than a touchpad
- gestures shortcuts are easier to remember than keyboard shortcuts

# Hypothesis

“Hypothesis is a formal statement that presents the expected relationship between an independent and dependent variable.”(Creswell, 1994)



A hypothesis should be:

- testable: the means for manipulating the variables and/or measuring the outcome variable must exist
- falsifiable: must be able to disprove the hypothesis with data
- precise: should be specific (operationalized)

# Testing a hypothesis

research hypothesis: Users point faster with a mouse than with a trackpad

null hypothesis: Users point as fast with a mouse as they point with a trackpad

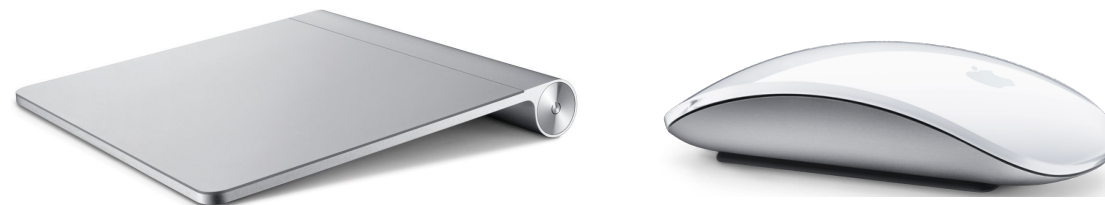
## Independent variable

Pointing device  
{Mouse,Trackpad}

leads to equal

## Dependent variable

Pointing time



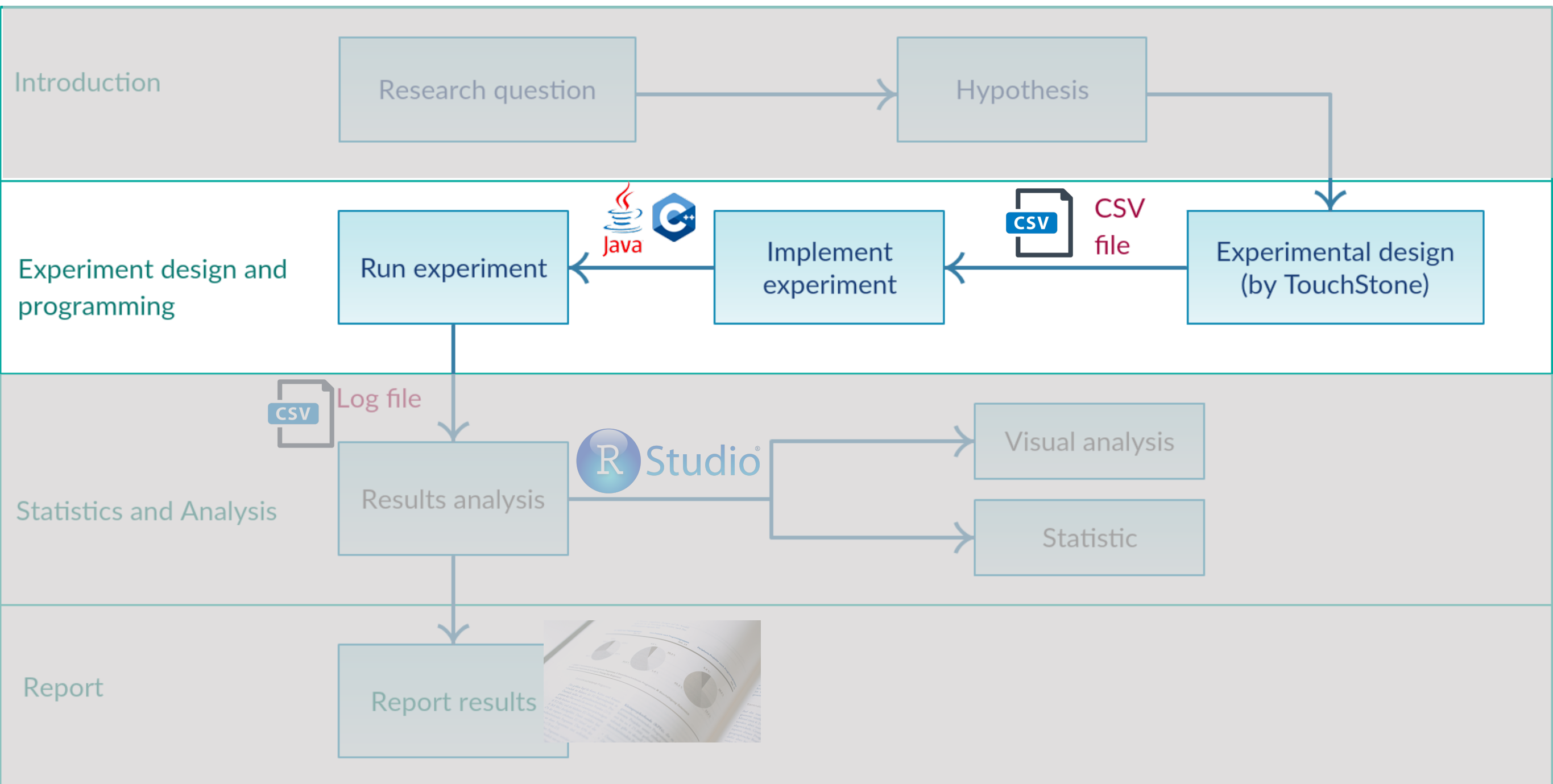
# Hypothesis

## Variables

- Independent: are varied under your control
  - E.g., number of menu entries
- Dependent: are measured
  - E.g., execution time, error rates, recall, subjective preferences, etc.

Hypothesis claims that **changing independent variables influences dependent variables**

- Example: Changing small to capital letters (independent variable) influences reading time (dependent variable)




[ABOUT US](#)
[HELP](#)
[FEEDBACK](#)

[HOME](#)
[Send SOME FLOOZ](#)
[Spend YOUR FLOOZ](#)
[ACCOUNT](#)

[SPECIALS](#)
[WHAT IS FLOOZ?](#)
[SECURITY GUARANTEE](#)



# The online gift currency that's thoughtful to send anytime, and fun to spend every time.

What does *flooz* mean? Meet *Fez*, your gift genie; he'll explain it all. >>>

**Security Guarantee**  
flooz is 100% secure. Read about our security commitment to you.



## flooz nooz

**NEW THIS WEEK!**  
**Gap Joins flooz**  
Gap joins the flooz crew. Another great place to shop with flooz.

[FLOOZ MERCHANT LIST >>>](#)

**FATHER'S DAY DILEMMA**  
**A Tie Rack? (yawn)**

With flooz, email Dad a groovy card, then let him choose the gift he really wants.

[SEE ALL OUR CARDS >>>](#)


[about](#)
[feedback](#)
[help!](#)

[HOME](#)
[send flooz](#)
[shop with flooz](#)
[ACCOUNT](#)

# The better way to gift!



*Flooz is the online gift certificate that lets receivers choose the perfect gift everytime.*

[meet Dot!](#)

**Father's Day is tomorrow!**  
but don't worry – there's still time to send flooz!

[choose a card to send](#)

Never miss an event or occasion again!

[register for reminders](#)



**IN THE NOOZ:**  
**The Gap is floozed!**

Cool fashions with flooz-friendly gifting now available!

[view merchants](#)

**GIFTAMONIAL:**  
I forgot my mother's birthday! It was a disaster. When I looked in my pilot, it was already too late to send a gift. But...

[submit your own giftamonal](#)

**THE STORY OF FLOOZ:**  
Ages ago, traders in the Mesopotamian valley used the word "Flooz" as slang for "cash" in the markets of ancient Persia.

[read the full Flooz story](#)

# Between groups



# Between groups



**flooz**  
The better way to gift.

HOME Send SOME FLOOZ Spend YOUR FLOOZ ACCOUNT  
SPECIALS WHAT IS FLOOZ? SECURITY GUARANTEE

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MasterCard American Express VISA TRUSTe

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about feedback help! HOME send flooz shop with flooz ACCOUNT

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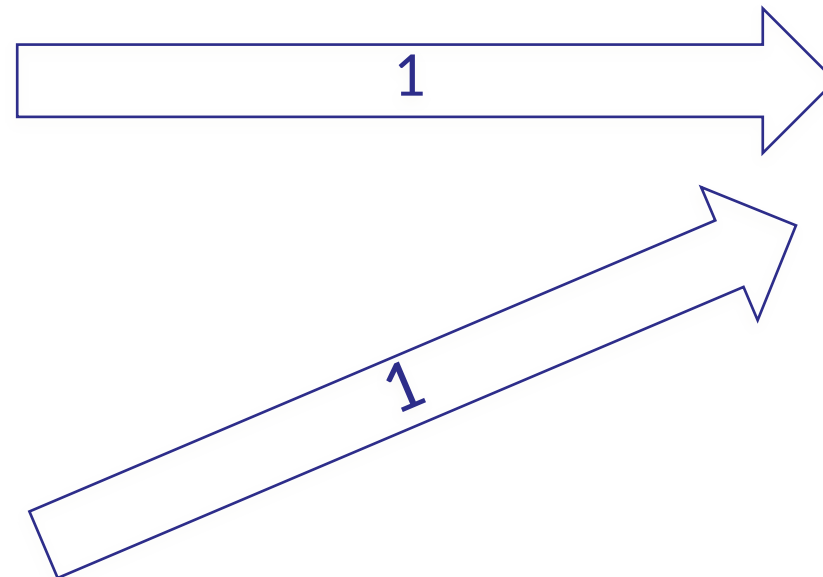
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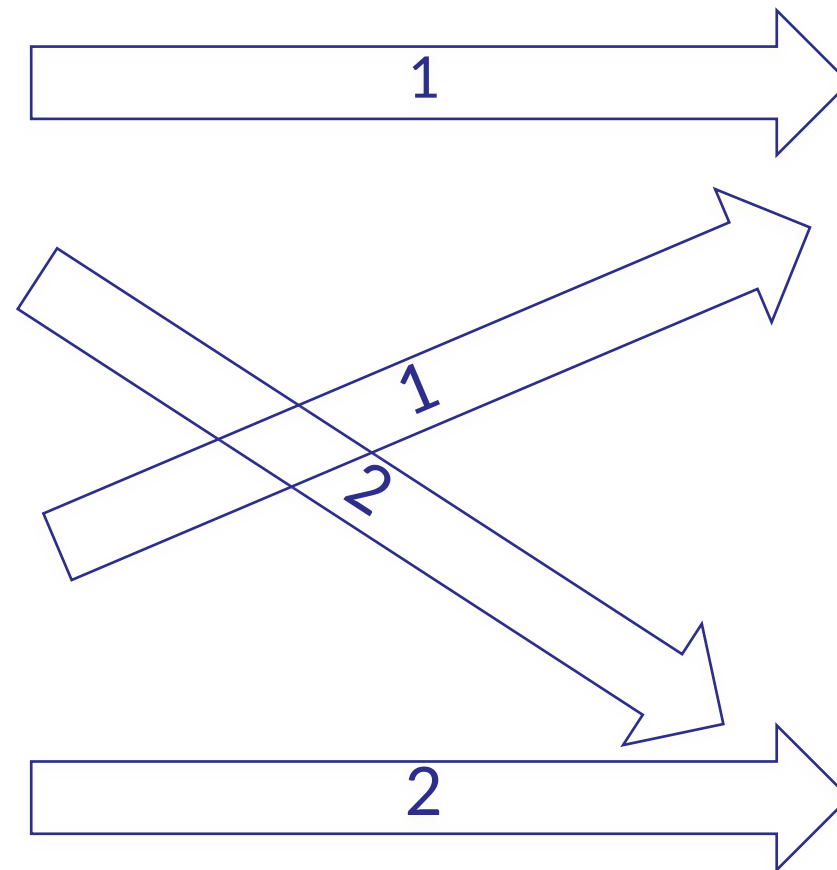
MasterCard American Express VISA VeriSign

# Within groups



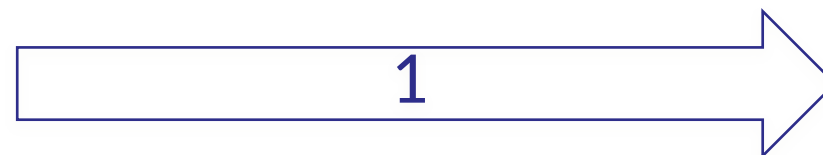
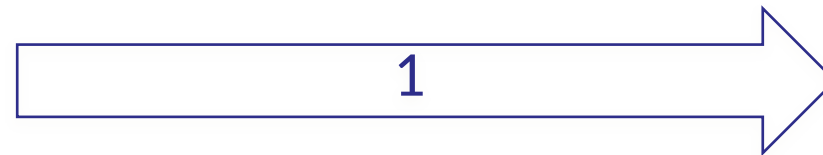
Learning effect.....

# Within groups

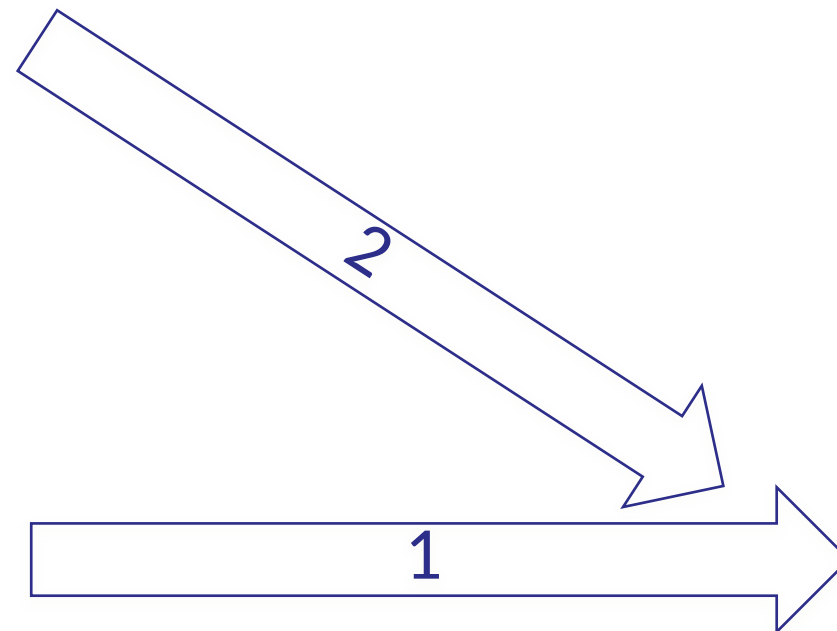
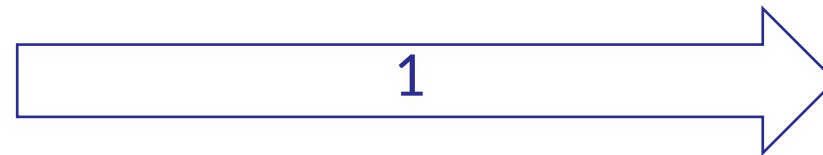


Learning effect.....

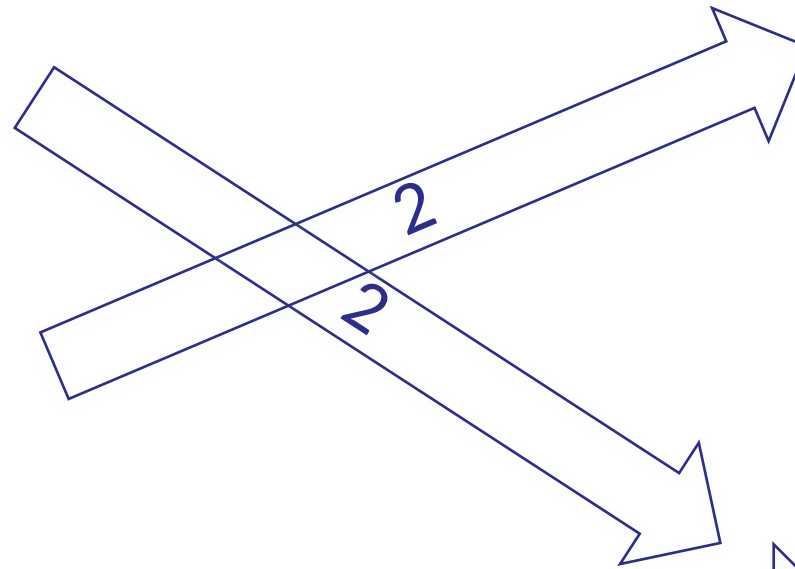
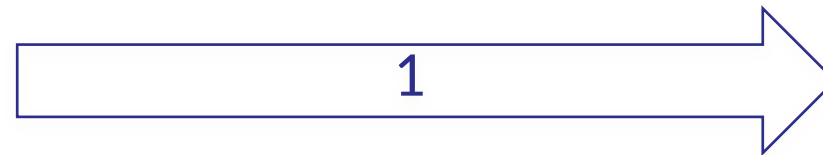
# Within groups

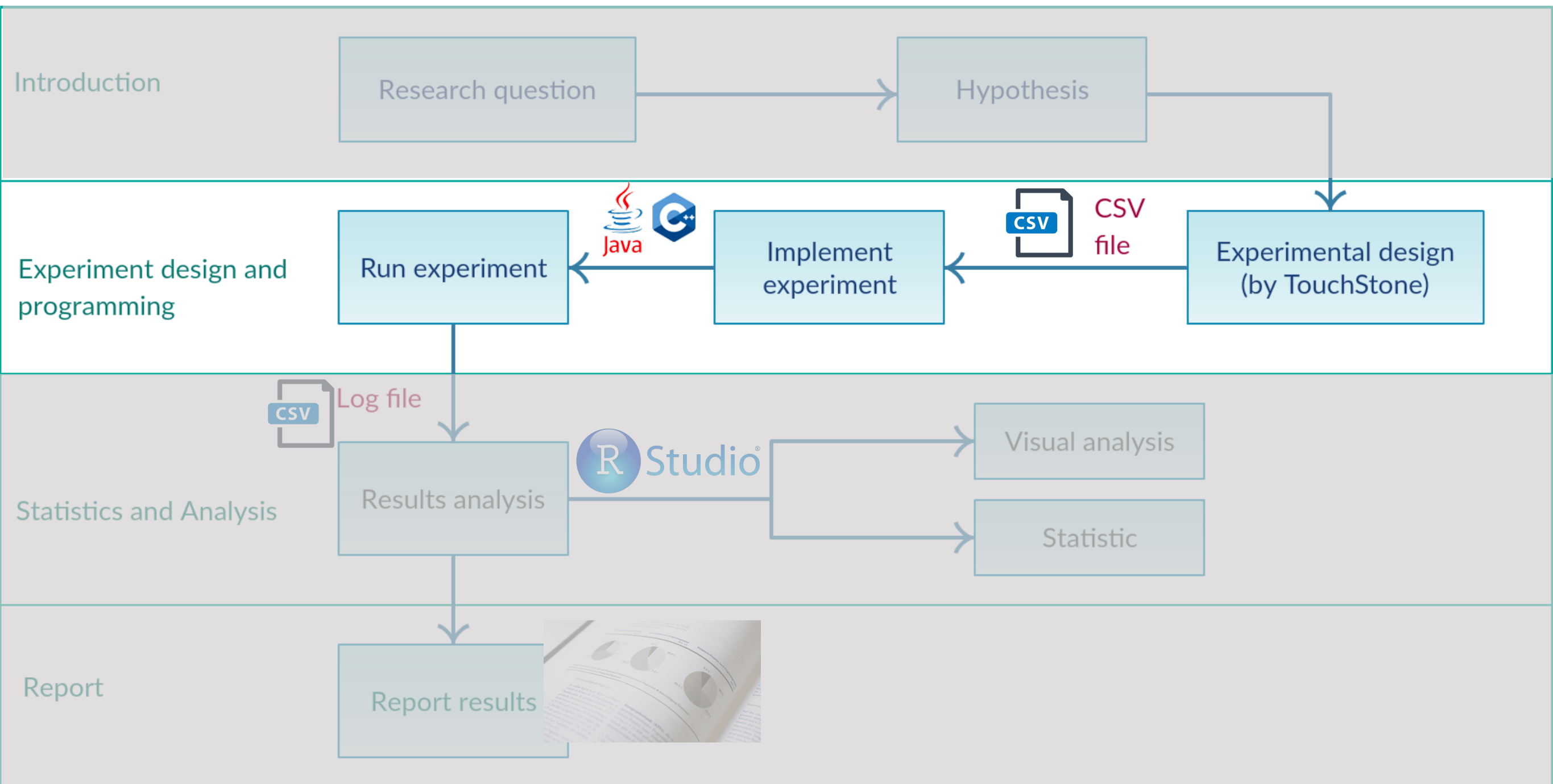


# Within groups



# Within groups





# CSV design description

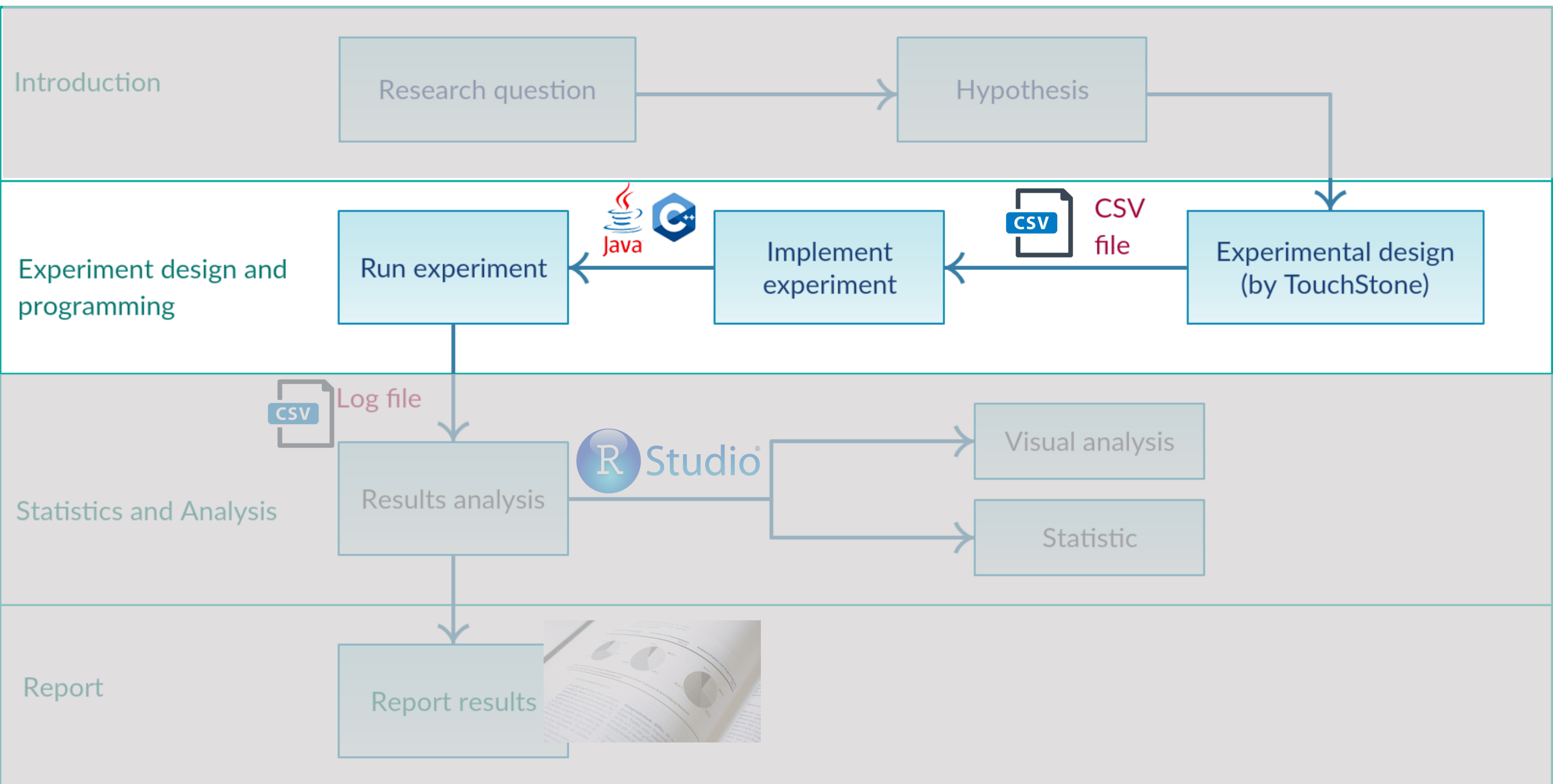
| Participant | Site | Task       | Block | Trial |
|-------------|------|------------|-------|-------|
| 1           | A    | navigation | 1     | 1     |
| 1           | A    | selection  | 1     | 2     |
| 1           | A    | navigation | 2     | 3     |
| 1           | A    | selection  | 2     | 4     |
| 2           | B    | navigation | 1     | 1     |
| 2           | B    | selection  | 1     | 2     |
| 2           | B    | navigation | 2     | 3     |
| 2           | B    | selection  | 2     | 4     |

File  
experiment.csv

## CSV File

### Characteristics :

One line for each trial,  
Comma separated  
fields, ...



# Programming

## Program this experiment

- load the right sequence of trials to run
- program the experiment engine for presenting trials and responding to user input
- log measured data

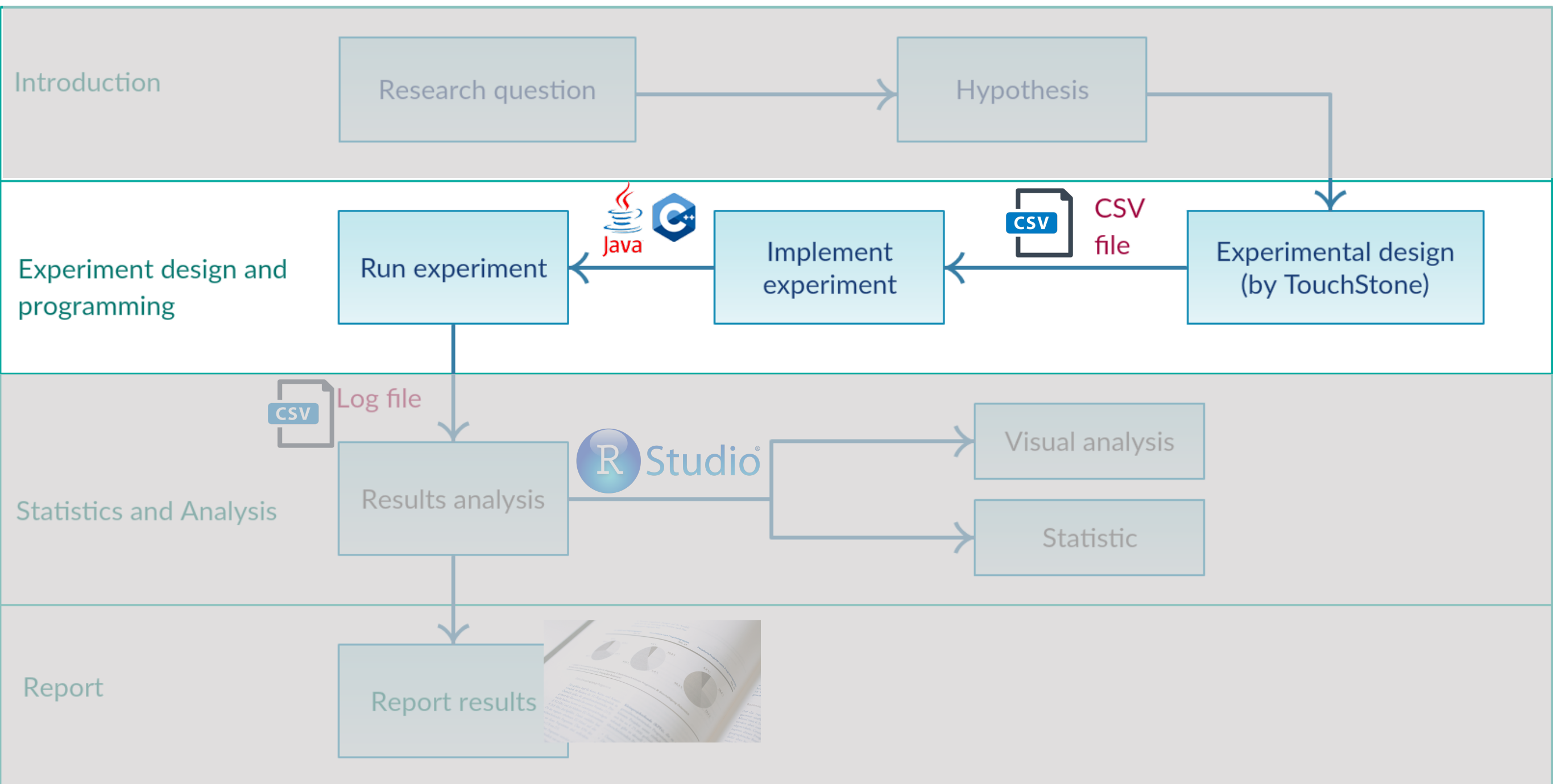
# CSV results

| Participant | Site | Task       | Block | Trial | Date | Time | Success | nb_fixation | .... |
|-------------|------|------------|-------|-------|------|------|---------|-------------|------|
| 1           | A    | navigation | 1     | 1     |      |      |         |             |      |
| 1           | A    | selection  | 1     | 2     |      |      |         |             |      |
| 1           | A    | navigation | 2     | 3     |      |      |         |             |      |
| 1           | A    | selection  | 2     | 4     |      |      |         |             |      |
| 2           | B    | navigation | 1     | 1     |      |      |         |             |      |
| 2           | B    | selection  | 1     | 2     |      |      |         |             |      |
| 2           | B    | navigation | 2     | 3     |      |      |         |             |      |
| 2           | B    | selection  | 2     | 4     |      |      |         |             |      |
| ...         |      |            |       |       |      |      |         |             |      |

## CSV File

### Characteristics :

One line for each record, Comma separated fields, ...



# Running the experiment

Consent form

Instructions for participants

Todo list for the experimenter

Observations vs. Non observations

Breaks

Debriefing

Questionnaire

Semi-structured interviews

# FEUILLE DE CONSENTEMENT

Bienvenue au Centre Multidisciplinaire des Sciences Comportementales Sorbonne Universités-INSEAD. Le chercheur de l'étude à laquelle vous allez participer est: Gilles BAILLY ([gilles.bailly@upmc.fr](mailto:gilles.bailly@upmc.fr)).

Dans cette étude en Interaction Homme-Machine (IHM), vous devrez sélectionner des commandes dans une interface de bureautique traditionnelle (écran, clavier, souris) aussi précisément et rapidement que possible. Le temps estimé nécessaire pour finir l'étude est de : 60 minutes.

Nos études sont à visées académiques, et les résultats seront dévoilés dans des publications scientifiques. Nous ne réalisons pas d'études pour le compte d'entreprises privées. Il n'existe aucun risque lié à cette étude autre que ceux de la vie de tous les jours.

Si vous terminez cette étude, vous recevrez 8 euros pour votre participation. Les trois participants avec les meilleurs temps recevront une prime de performance de 20 euros pour le premier, 15 euros pour le second et 10 euros pour le troisième.

Par ailleurs, vous pourrez recevoir des informations sur les conclusions de l'étude si vous le souhaitez, ainsi que des références concernant le type de recherche auquel vous avez participé. Cependant, parce que vos réponses sont anonymes, nous ne pourrions vous renseigner que sur les résultats agrégés de l'étude, et non sur votre performance ou les performances de n'importe qui d'autre ayant participé.

Nous aimerions également votre permission de procéder à un enregistrement vidéo/audio de l'étude. Si nous avons besoin d'utiliser des citations ou des éléments susceptibles de vous identifier, nous vous demanderons alors votre permission avant de le

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Les données concernant cette étude incluant les enregistrements seront conservées sous clé ou protégées par un mot de passe, et seront détruites dès lors qu'elles ne seront plus utilisées.

Votre participation à l'étude doit être entièrement volontaire, et vous avez la possibilité de vous retirer de l'étude à tout moment sans aucune pénalité.

Je déclare être majeur(e), et ayant lu et parfaitement compris les paragraphes ci-dessus, accepte de mon plein gré de participer à cette étude.

DATE:

NOM:

SIGNATURE:

## **Explication de ce type de recherche**

Cette étude s'inscrit dans le domaine de l'Interaction Homme-Machine (IHM), la science qui étudie les interactions entre un ou plusieurs utilisateurs et un ou plusieurs systèmes interactifs (ordinateurs, tablettes, téléphones portables, etc.). Cette étude se concentre sur les interfaces reposant sur des systèmes de menus comme on peut en trouver dans la plupart des applications de bureautique. Bien que ces interfaces soient largement utilisées, elles ont été relativement peu étudiées.

## **Description plus détaillée de l'étude**

Dans cette étude nous souhaitons mieux comprendre comment les utilisateurs sélectionnent les commandes, est ce que des stratégies émergent en fonction des commandes et des participants et est ce que ces stratégies évoluent au cours de du temps. L'objectif à moyen terme est de développer de nouvelles formes d'interaction pour la sélection de commandes qui améliorent la productivité au jour le jour.

Vous pouvez trouver plus d 'information sur ce sujet dans l'ouvrage ci-dessous :

- Gilles Bailly, Eric Lecolinet, and Laurence Nigay. 2016. Visual Menu Techniques. *ACM Comput. Surv.* 49, 4, Article 60 (December 2016), 41 pages. <https://hal.archives-ouvertes.fr/hal-01258368v2>

Si vous avez des questions concernant cette étude, vous pouvez nous contacter :

Gilles BAILLY : [gilles.bailly@upmc.fr](mailto:gilles.bailly@upmc.fr)

## **A propos du Centre Sorbonne Universités-INSEAD**

Le Centre Sorbonne Universités-INSEAD est dédié à la recherche comportementale.

# Running the experiment

Ensure that subjects remain anonymous

- Associate a number with each subject
- Choose conditions based on those numbers

# Running the experiment

Consent form

Instructions for participants

Todo list for the experimenter

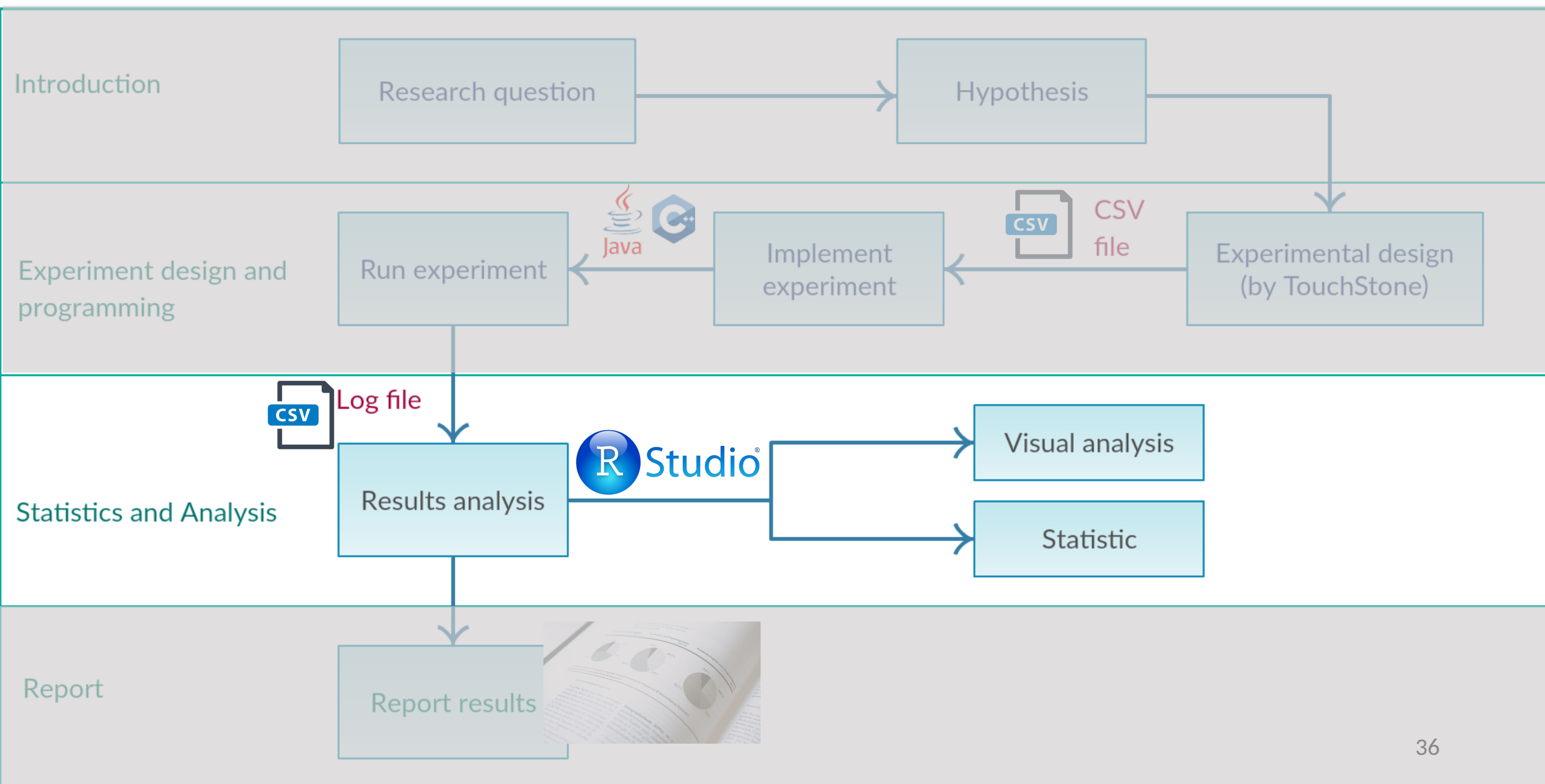
Observations vs. Non observations

Breaks

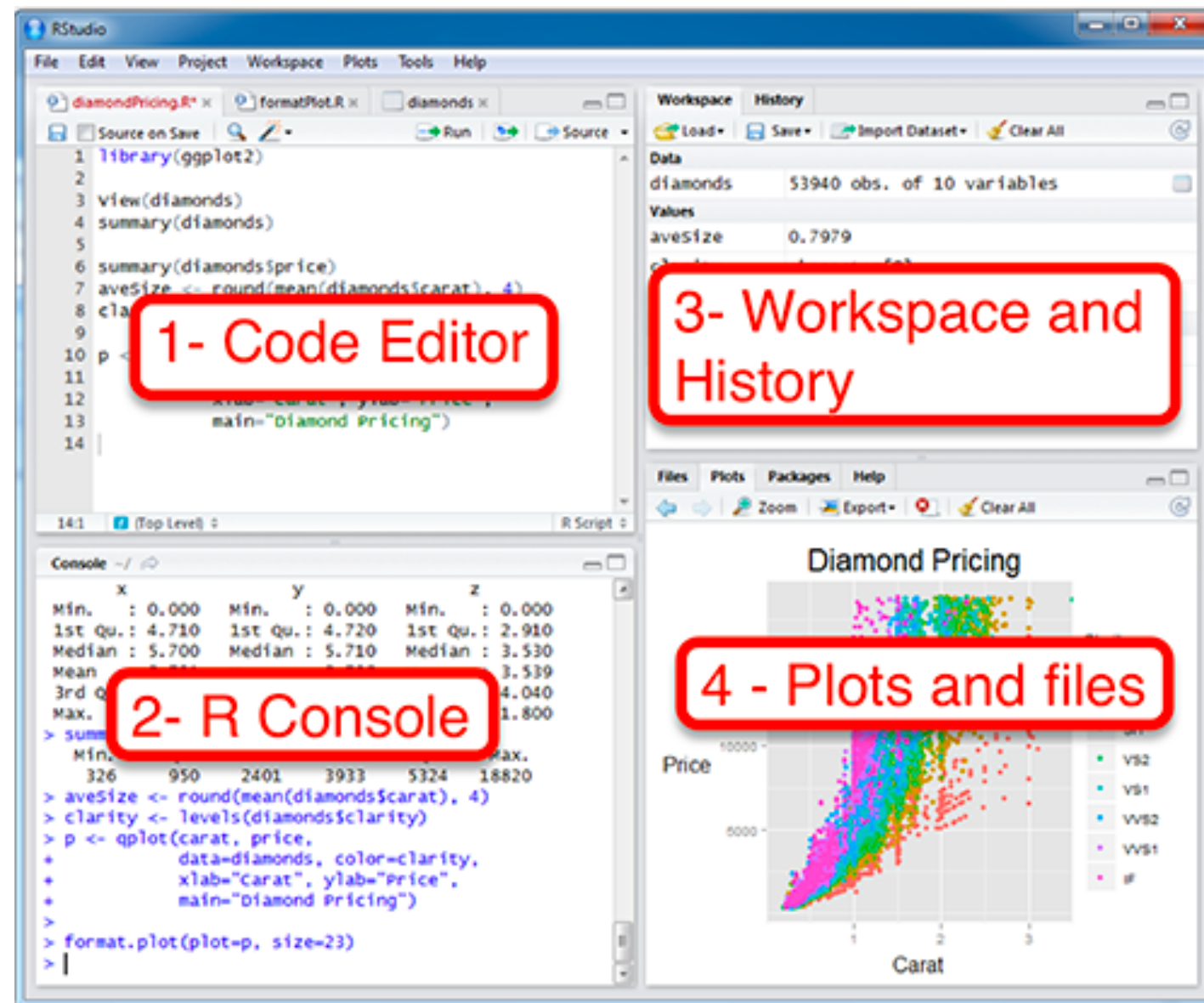
Debriefing

Questionnaire

Semi-structured interviews

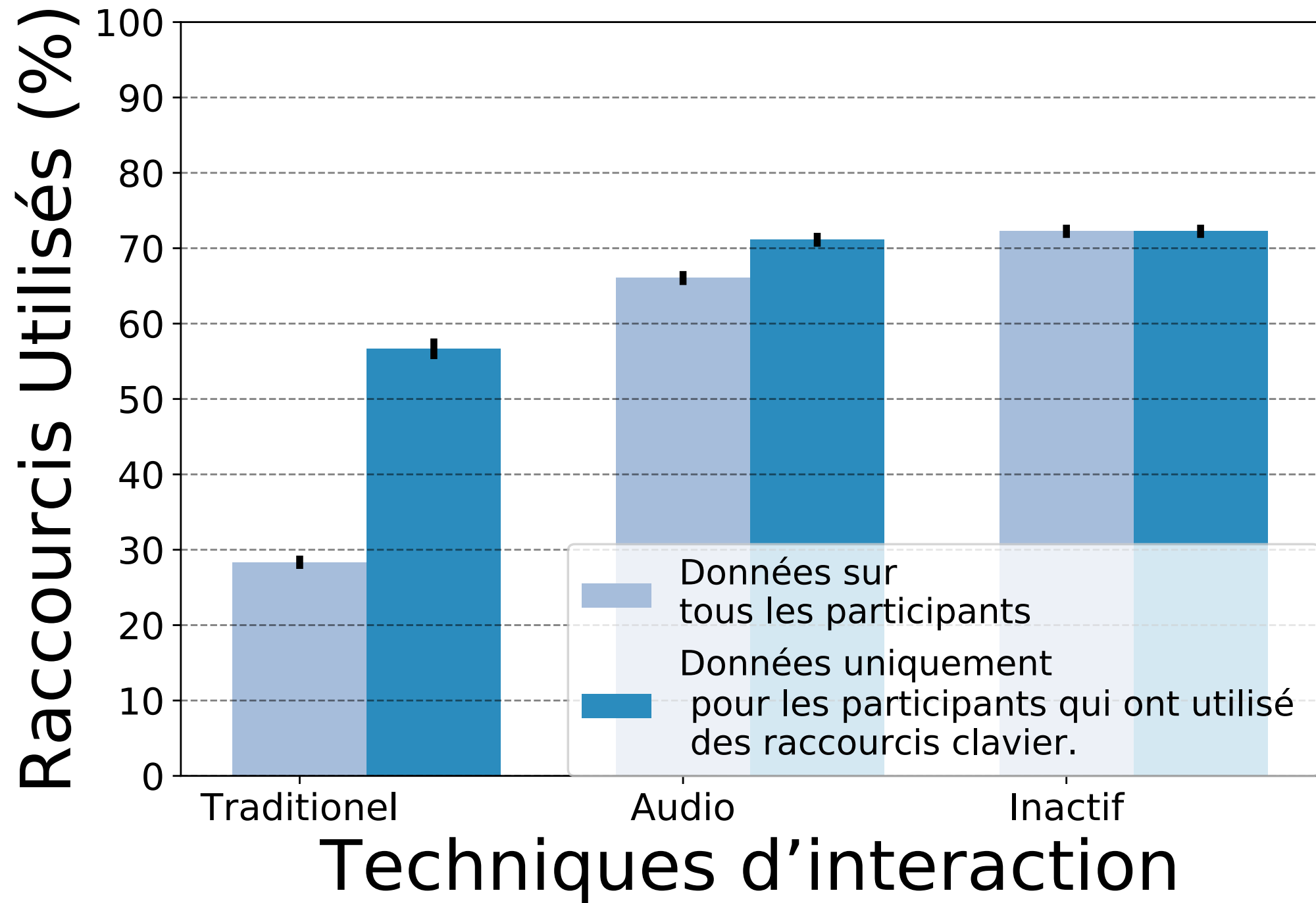


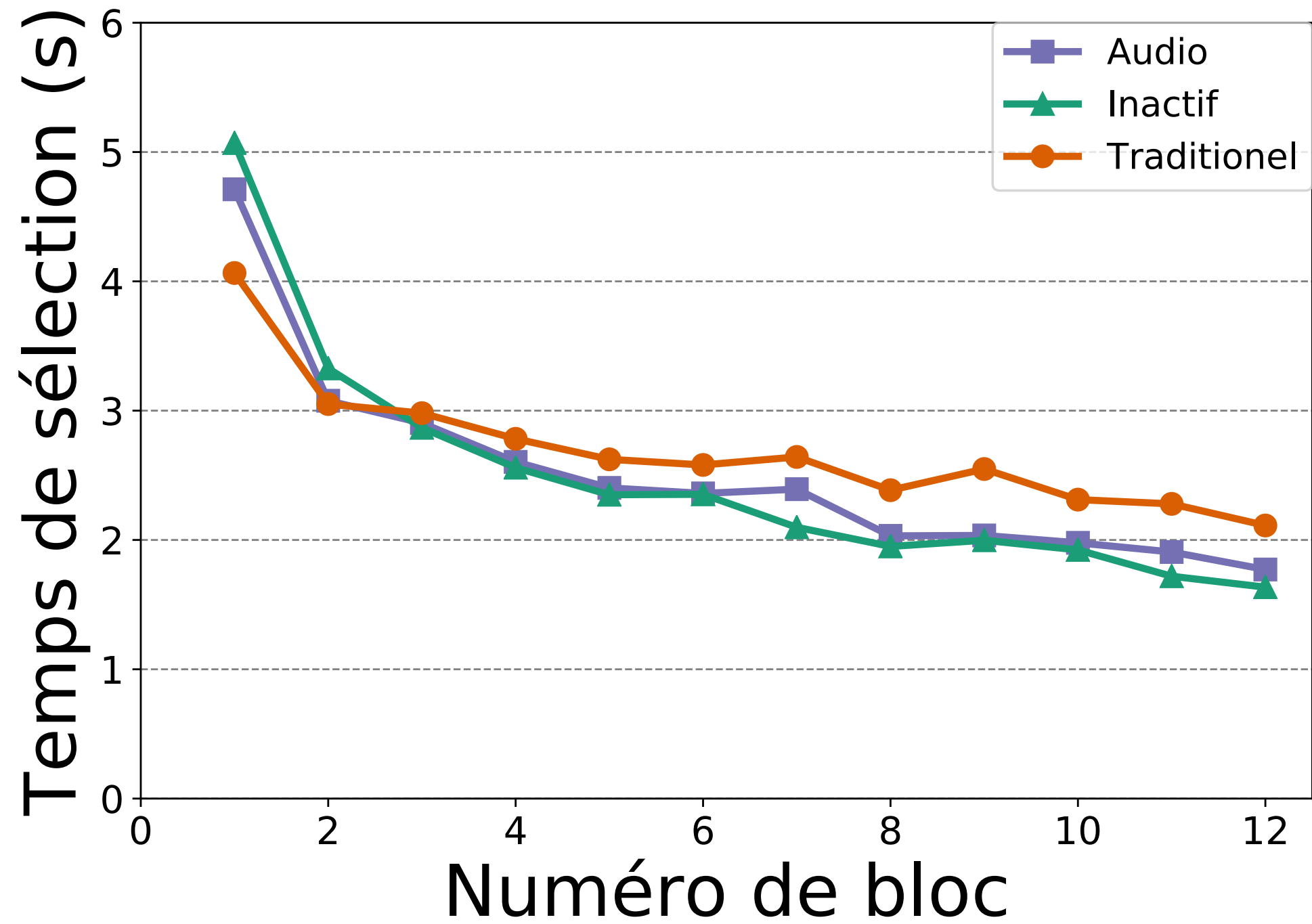
# R Studio



Statistical  
analysis

Explore the  
result visually



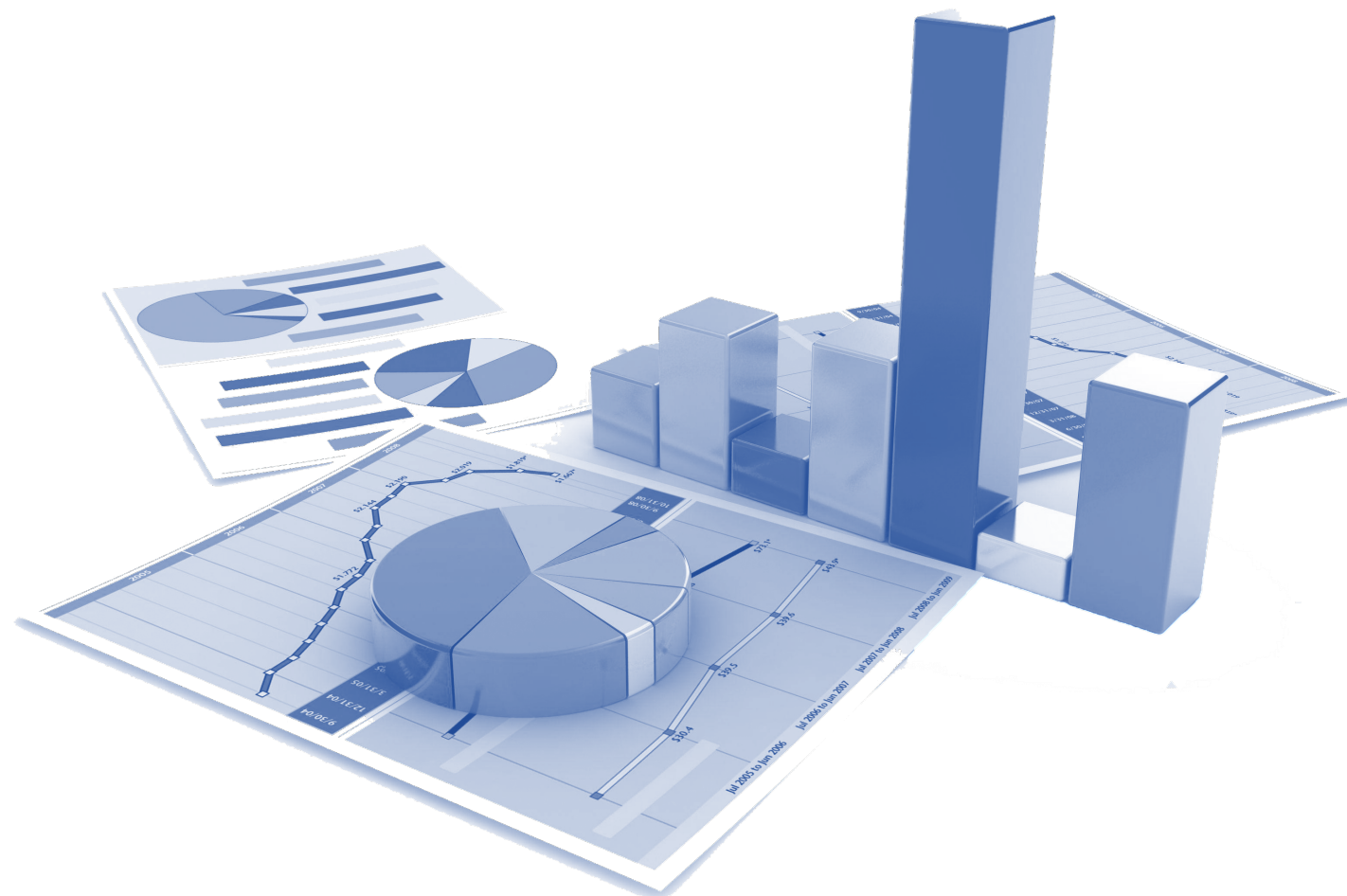


# Statistical analysis

Provides the mathematical characteristics of data

Describes how data sets are related to each other

Estimates the probability that hypothesis are correct



# Statistical tests in R

Chi-Square/McNemar

t-test

One-way ANOVA

Two-way ANOVA

Linear regression

# Statistical tests in R

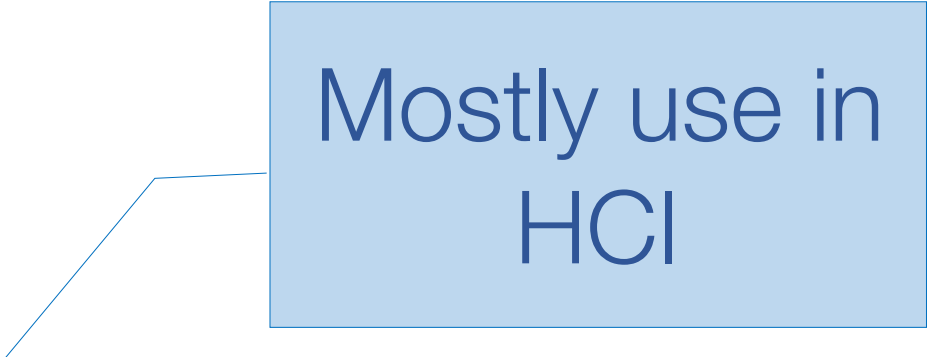
Chi-Square/McNemar

t-test

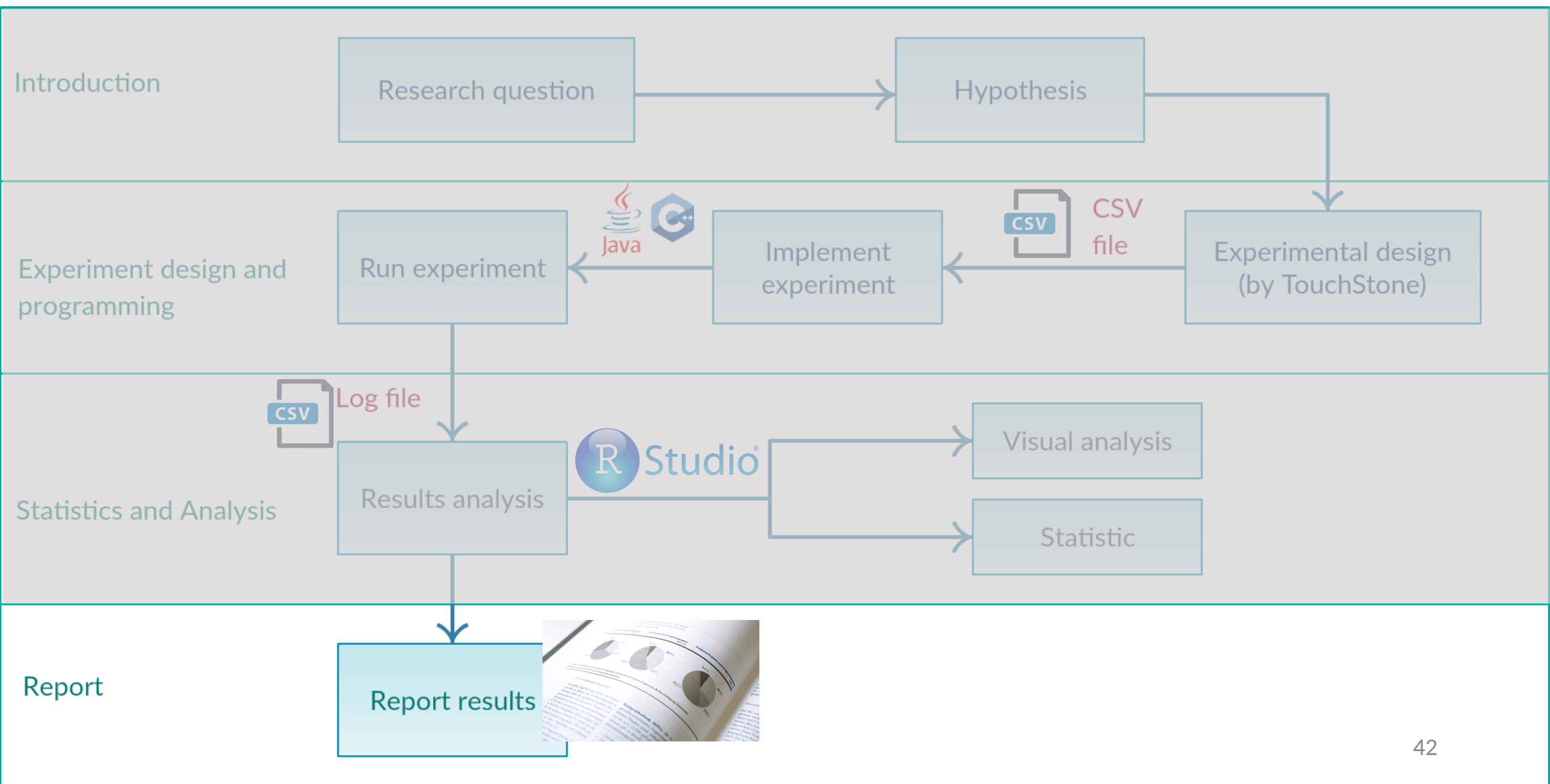
One-way ANOVA

Two-way ANOVA

Linear regression



Mostly use in  
HCI



available software and Newton-X.

and related extension method, finding the language environment with natural systems made by them when the trajectories had been proposed ended the states.<sup>12</sup>

and the  
treatment  
and the  
the TSH  
it has  
the 199  
it is ch-

...ing Credit B Co. REBA, Washington



and the mean-field approach also treatment.<sup>35</sup> This concept of mix- and the TSH approaches has the TSH from *ad hoc* abrupt trans- it has still been explored in recent (the 1990s, the exact

$$A_{j,\alpha\beta}(\mathbf{R}) = -\frac{1}{\Omega_j(\mathbf{R})} \frac{\text{Tr}[\ell_j^+(\mathbf{R}) A_{\alpha\beta}(\mathbf{R})]}{(1 - \text{Tr}[\rho(\mathbf{R})[\ell_j^+(\mathbf{R}), L^*(\Omega_j, \mathbf{R}) \ell_j(\mathbf{R})]])^{1/2}} \quad (\text{C14})$$
[illegible]
$$A_{j,\mu}(\mathbf{R}) = \frac{1}{\Omega_j(\mathbf{R})} \frac{\text{Tr}[\hat{\ell}_j^{\dagger}(\mathbf{R}) \hat{h}_{\mu}(\mathbf{R})]}{(1 - \text{Tr}[\rho(\mathbf{R})[\hat{\ell}_j^{\dagger}(\mathbf{R}), L'(\Omega_j, \mathbf{R})\hat{\ell}_j(\mathbf{R})])^{1/2}} \quad (\text{C14})$$

These are not the final page numbers!

# Report structure

## Introduction

- Goal
- Hypotheses

Participants (age, sexe, skills, etc.)

Apparatus (devices, setup, etc.)

## Methods

- Stimulus and Task
- Procedure (instructions, pre-test, post-test, questionnaire, etc.)
- Independent and dependent variables
- Design (within, between, block, repetition, etc.)

## Results

- statistical analyses

## Discussion

- Revisit hypotheses
- what worked well, potential limits and threats of validity, etc.

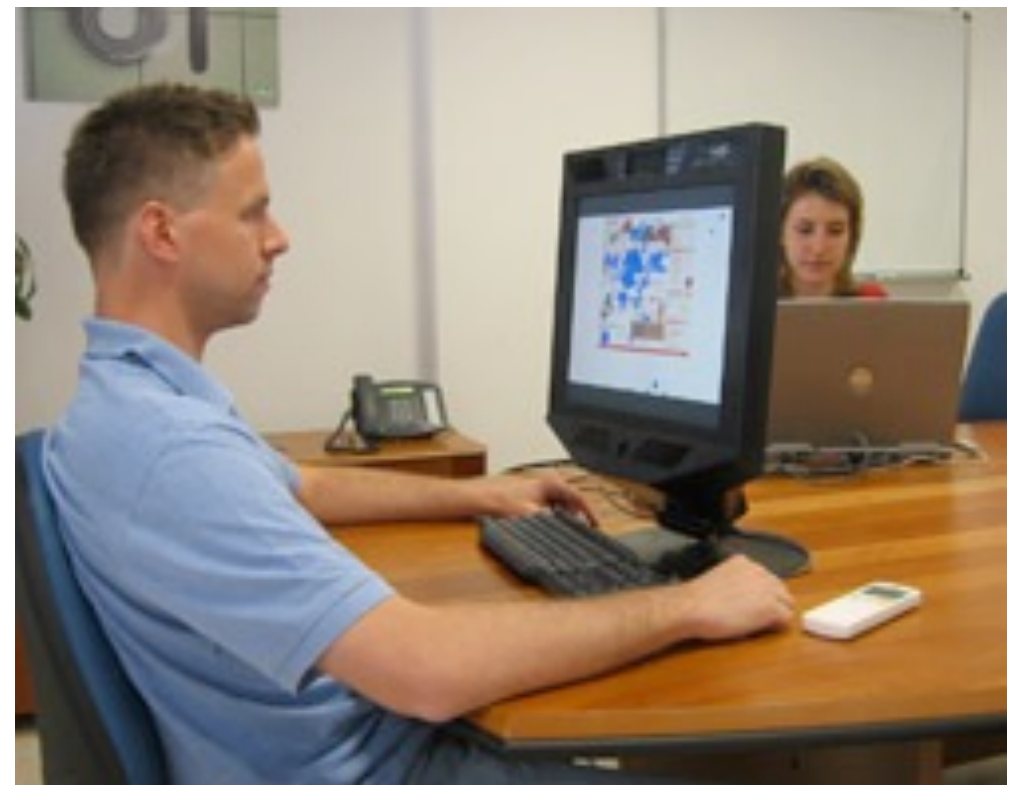
# Eye Tracking in Usability Tests

What users are looking at...

Detailed usability metrics while performing task

Analysis extremely time consuming

More details next week



Source: <http://www.uxmatters.com/MT/archives/000040.php>

# Fixations and Saccades while Reading

## DANS, KÖN OCH JAGPROJEKT

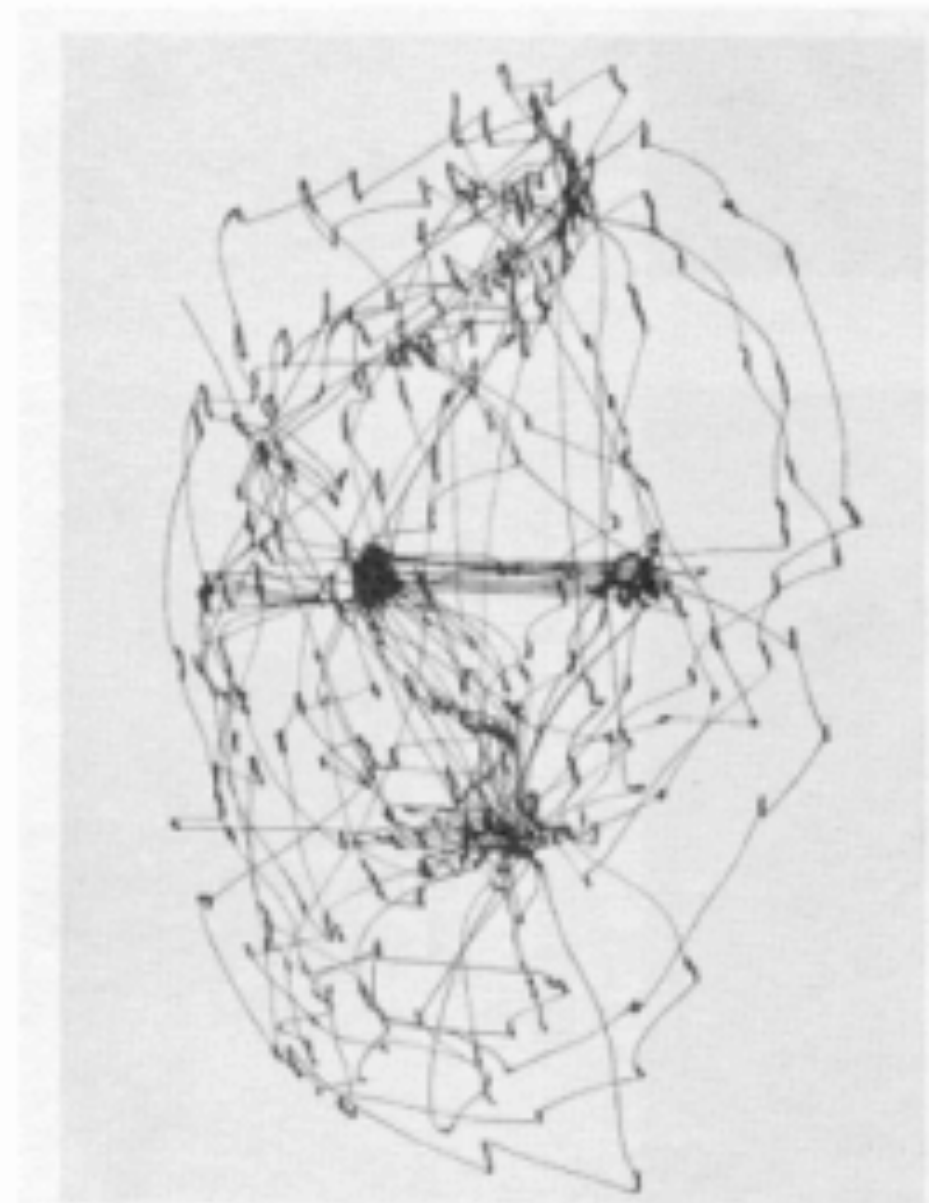
På jakt efter ungdomars kroppsspråk och den "synkretiska dansen", en sammansmältning av olika kulturellers dans, har jag i mitt fältarbete under hösten rört mig på olika arenor inom skolans värld. Nordiska, afrikanska, syd- och östeuropeiska ungdomar gör sina röster hörda genom sång, musik, skrik, skratt och gestaltar känslor och uttryck med hjälp av kroppsspråk och dans.

Den individuella estetiken framträder i kläder, frisyrer och symboliska tecken som förstärker ungdomarnas "jagprojekt" där också den egna stilen i kroppsrörelserna spelar en betydande roll i identitetsprövningen. Upphållsrummet fungerar som offentlig arena där ungdomarna spelar upp sina performanceliknande kroppsshower

# Eye Movements for Images

Fixation of information bearing structures

- For faces: eyes and mouth



Source: [http://www.paed.uni-muenchen.de/~deubel/Motorik/Integrat\\_web/ppframe.htm](http://www.paed.uni-muenchen.de/~deubel/Motorik/Integrat_web/ppframe.htm)

Wake up  
the red  
in your hair.



SUNSILK

Wake up  
the red  
in your hair.



SUNSILK



[ Source : James Hudson, PayPal ]

Before



Extra gentle for the  
most sensitive skin.

Start with ultra sensitive skin, add the chemicals and moisture  
of urine and stools, and you have diaper rash.



After



Extra gentle for the  
most sensitive skin.

Start with ultra sensitive skin, add the chemicals and moisture  
of urine and stools, and you have diaper rash.



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PayPal password

Go to

My account 

[Log In](#)

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New to PayPal? [Sign up](#).

## Top questions

- [Why use PayPal when I have credit cards?](#)
- [What can I do with PayPal?](#)
- [Is PayPal free to use?](#)



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Home

Personal

Business

Products &amp; Services

Shopping

Get Started

Send Money

Request Money

Sell on eBay

Integrate

## Account login

Email address

PayPal password

Go to

Forgot your [email address](#) or [password](#)?New to PayPal? [Sign up](#).

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- [Why use PayPal when I have credit cards?](#)
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Your Zip Code:

Email:

We value your privacy and will not sell or give your information to anyone. Period.

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*Simple  
&  
Easy*



Le Laboratoire des Usages en Technologies d'Information Numériques ( LUTIN - Dir.: Charles Tijus; Dir. Scientifique: Thierry Baccino), est un [LIVING LAB](#) européen des Universités Paris 8, Paris 6, UTCompiègne et de Universcience, dédié à l'étude des usages des nouvelles technologies dans le cadre de recherches académiques et industrielles.



## Actualités

### Signature d'un Memorandum of Understanding avec Hong Bang au Vietnam



Un Memorandum of Understanding (MOU) a été signé entre le LUTIN et la société Hong Bang, d'Hanoï, chargée de la promotion et du financement des nouvelles technologies au Vietnam. L'objectif de cet accord vise à un étroit partenariat entre les parties, portant sur le **développement d'un Living Lab**, nommé **Tinh Nghich**, situé à Hanoï et dédié aux tests d'usage et aux technologies de détection (sensors technology).

### Accord de coopération entre le LUTIN et l'Université Sungkyunkwan



Un accord de coopération a été signé entre le LUTIN et le département [Interaction Science](#) (IS) de l'[Université de Sungkyunkwan](#), située à Séoul, en Corée du Sud. Cet accord vise à renforcer la collaboration de recherche et d'enseignement entre les deux laboratoires. Ils décident mutuellement de promouvoir les échanges afin que

## Eye tracking Lab

8 Eye trackers permettent de comprendre ce que regarde l'utilisateur dans toutes les situations

## Game Room

Découvrez la plateforme et les méthodes d'analyse du jeu vidéo pour améliorer l'ergonomie et le gameplay des jeux en cours de développement

## Mobility Lab

Techniques et méthodes d'observation pour étudier et accompagner l'innovation dans les

# Video-Based Eye Trackers

## Mobile and stationary trackers

- Desktop monitor with integrated eye tracker
- Bike helmet with eyetracker and camera capturing user's field of view



infrared light

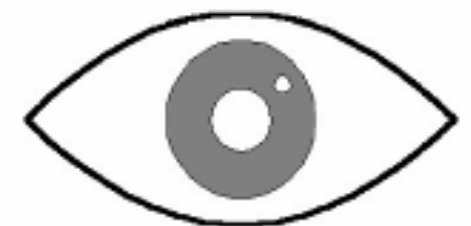
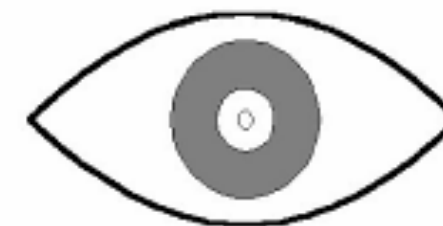
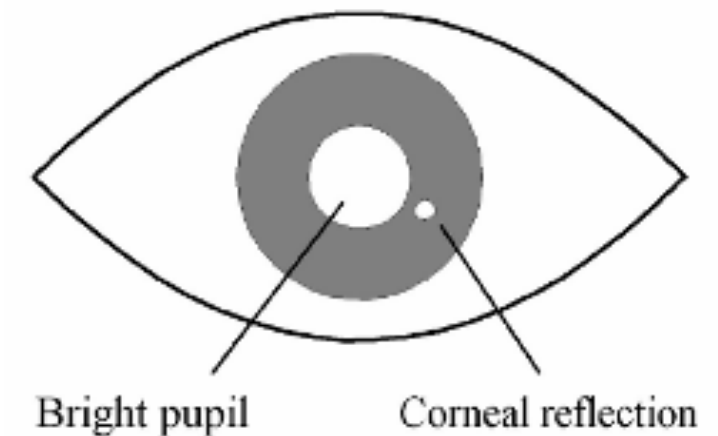


Source: <http://www.uxmatters.com/MT/archives/000040.php>

# How Video-Based Eye Trackers Work

## Infrared illumination of the eye

- On-axis IR light source: bright pupil, “red eye” effect
- Off-axis IR light source: dark pupil



## Bright spot from corneal reflection

Source: Morimoto et al.:  
Pupil detection and tracking  
using multiple light sources.  
Image and Vision  
Computing 18, 2000, pp.  
331-334.

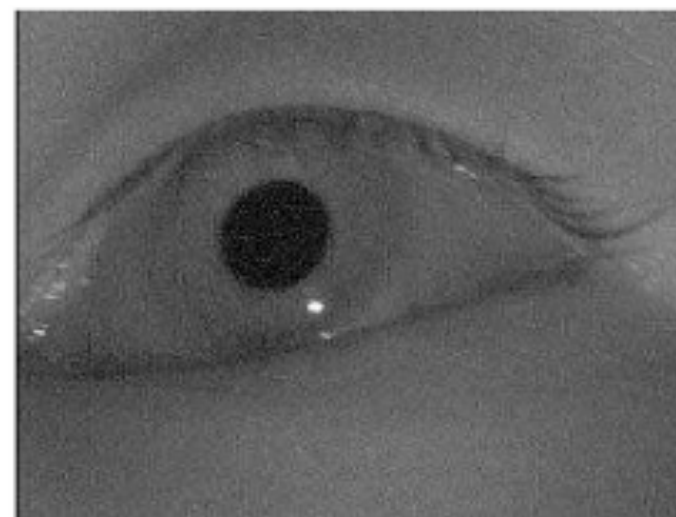


Figure 4: Dark pupil tracking illuminates the eye with an off-axis IR light source and extracts the black pupil [11].

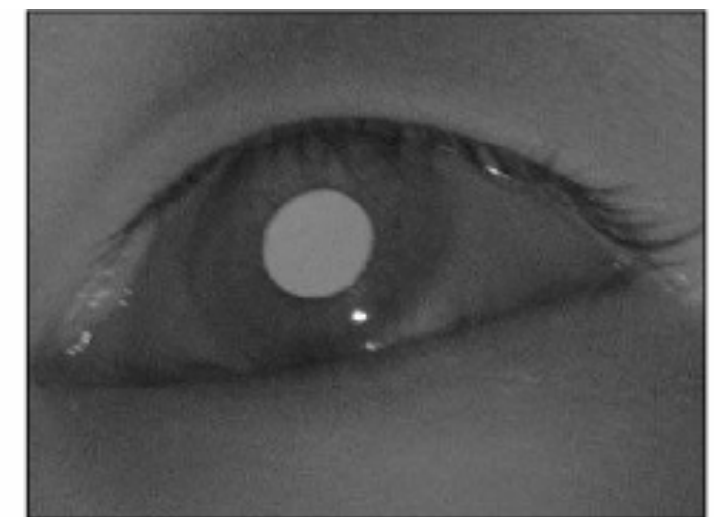
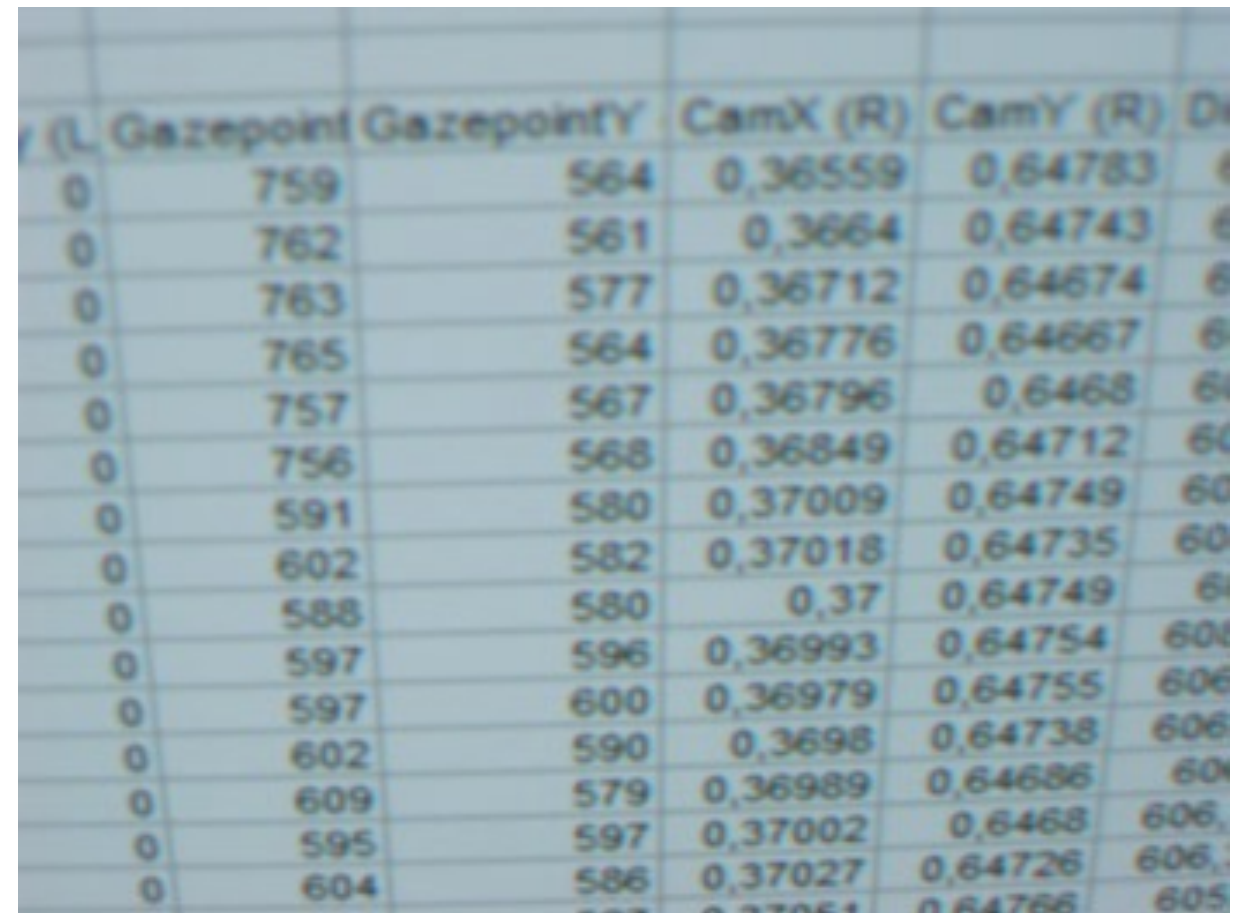


Figure 5: Bright pupil tracking uses an IR light source located on the visual axis of the camera to induce the “red eye” effect seen in flash photography [11].

# Eye Tracking Data

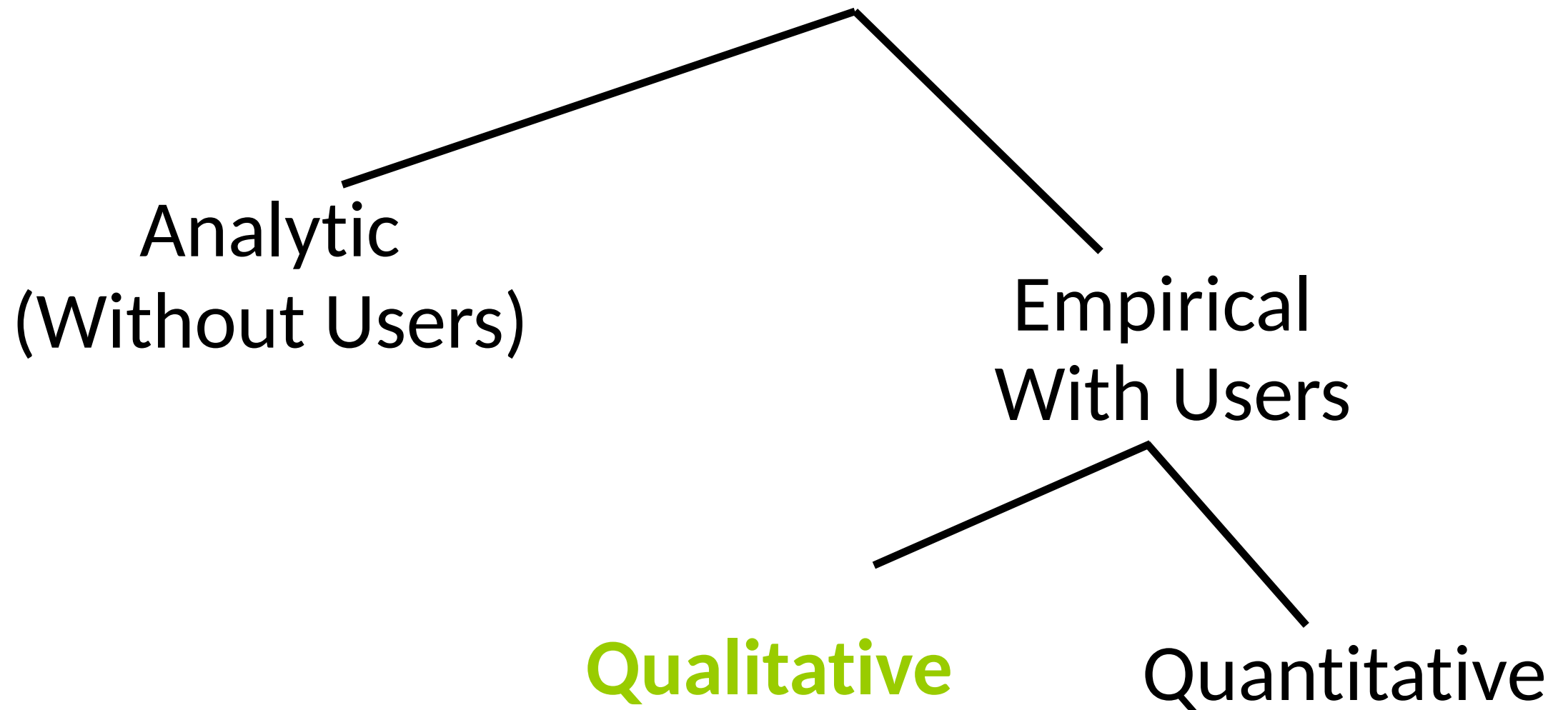
- Large quantity of raw data
  - Angle of saccadic movement
  - Fixation coordinates
  - Fixation duration
  - Pupil diameter
  - etc.
- Statistical analysis
- Visualization



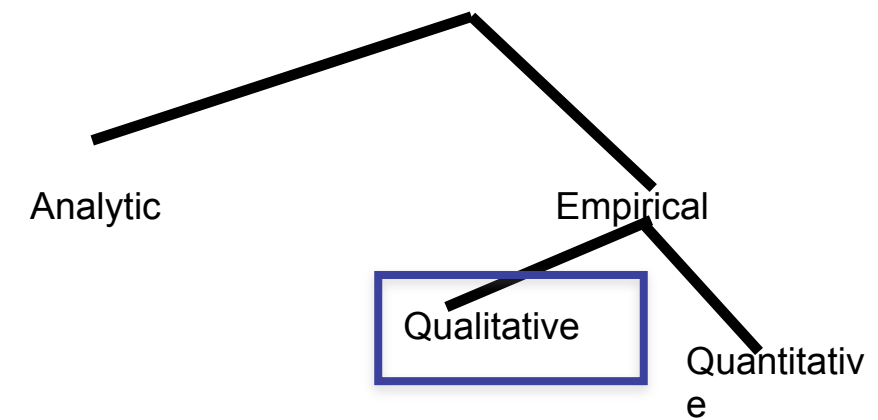
| L | Gazepoint | GazepointY | CamX (R) | CamY (R) | D  |
|---|-----------|------------|----------|----------|----|
| 0 | 759       | 564        | 0,36559  | 0,64783  | 6  |
| 0 | 762       | 561        | 0,3664   | 0,64743  | 6  |
| 0 | 763       | 577        | 0,36712  | 0,64674  | 6  |
| 0 | 765       | 564        | 0,36776  | 0,64667  | 6  |
| 0 | 757       | 567        | 0,36796  | 0,6468   | 6  |
| 0 | 756       | 568        | 0,36849  | 0,64712  | 60 |
| 0 | 591       | 580        | 0,37009  | 0,64749  | 60 |
| 0 | 602       | 582        | 0,37018  | 0,64735  | 60 |
| 0 | 588       | 580        | 0,37     | 0,64749  | 6  |
| 0 | 597       | 596        | 0,36993  | 0,64754  | 60 |
| 0 | 597       | 600        | 0,36979  | 0,64755  | 60 |
| 0 | 602       | 590        | 0,3698   | 0,64738  | 60 |
| 0 | 609       | 579        | 0,36989  | 0,64686  | 60 |
| 0 | 595       | 597        | 0,37002  | 0,6468   | 60 |
| 0 | 604       | 586        | 0,37027  | 0,64726  | 60 |
| 0 |           |            | 0,37051  | 0,64766  | 60 |

Source: <http://www.uxmatters.com/MT/archives/000040.php>

# Evaluation Methods



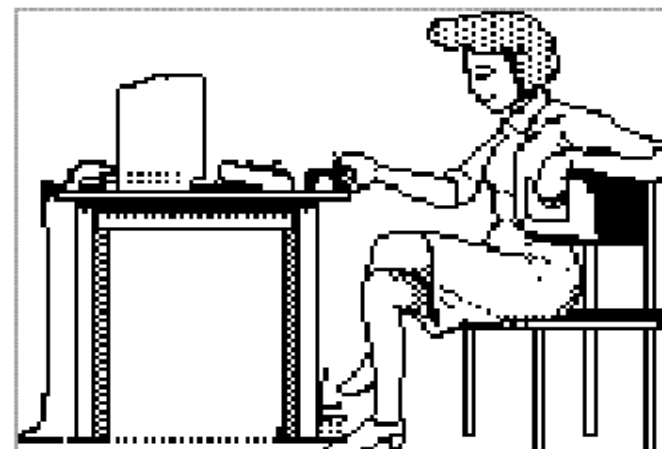
# Silent Observation

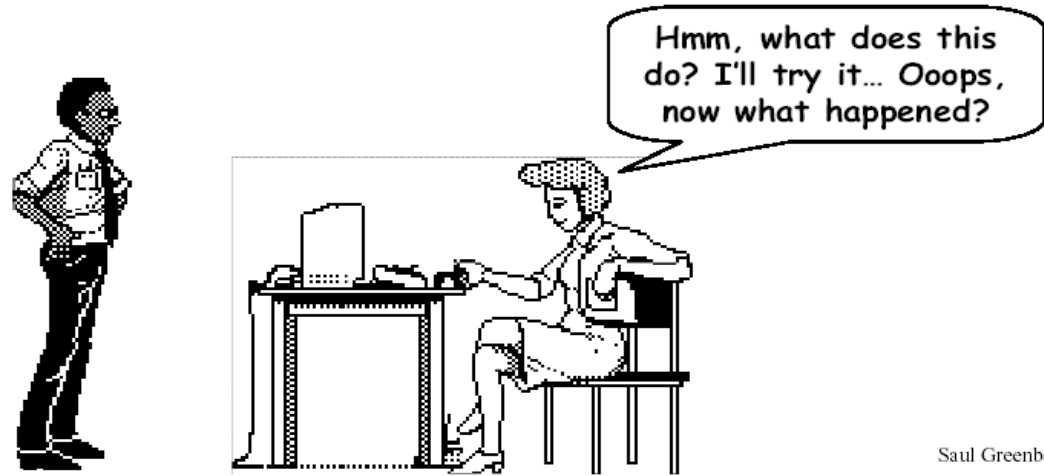


Designer **watches** user in lab or in natural environment while working on one of the tasks

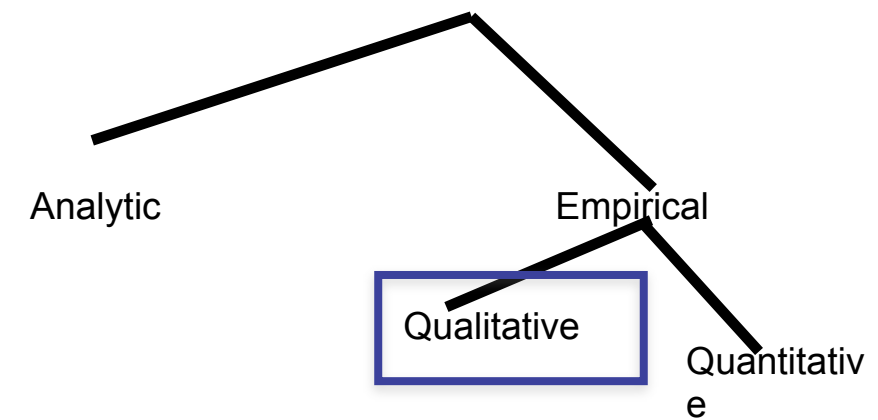
**No communication** during observation

- + Helps discover big problems
- No understanding of decision process (that may be wrong) or user's mental model, opinions, or feelings





rk Aloud



User is asked to **say aloud**

- **State**: What he thinks is happening.
- **Goals**: What he is trying to achieve
- **Actions**: Why he is doing something specific

Good to get some insight into user's thinking, but:

- Talking is hard while focusing on a task
- Feels weird for most users to talk aloud
- Conscious talking can change behavior



## WhizzGo.com Test participant: James.

from nicepaul 5 years ago NOT YET RATED



## WhizzGo.com Test participant: James.

from nicepaul 5 years ago NOT YET RATED



## WhizzGo.com Test participant: James.

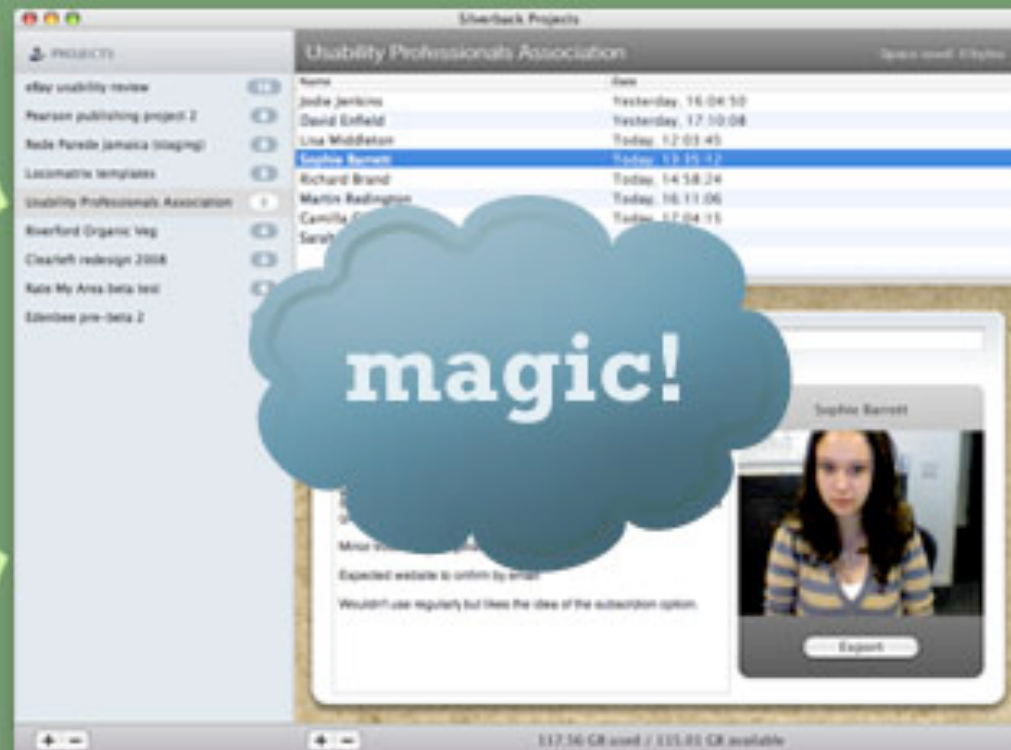
from nicepaul 5 years ago NOT YET RATED



screen capture



live video & audio



Silverback



Quicktime  
composite

# Describe

- What you see
- What you do
- What you expect to get



# Recording Observations

## Paper and pencil

- Evaluator notes events, interpretations, other observations
- Cheap but hard with many details (writing is slow)
- Forms can help

## Audio recording

- Good for speech with Think Aloud and Constructive Interaction
- But hard to connect to interface state

## Video

- Ideal: 2 cameras (user and screen) in 1 picture
- Best capture, but may be too intrusive initially

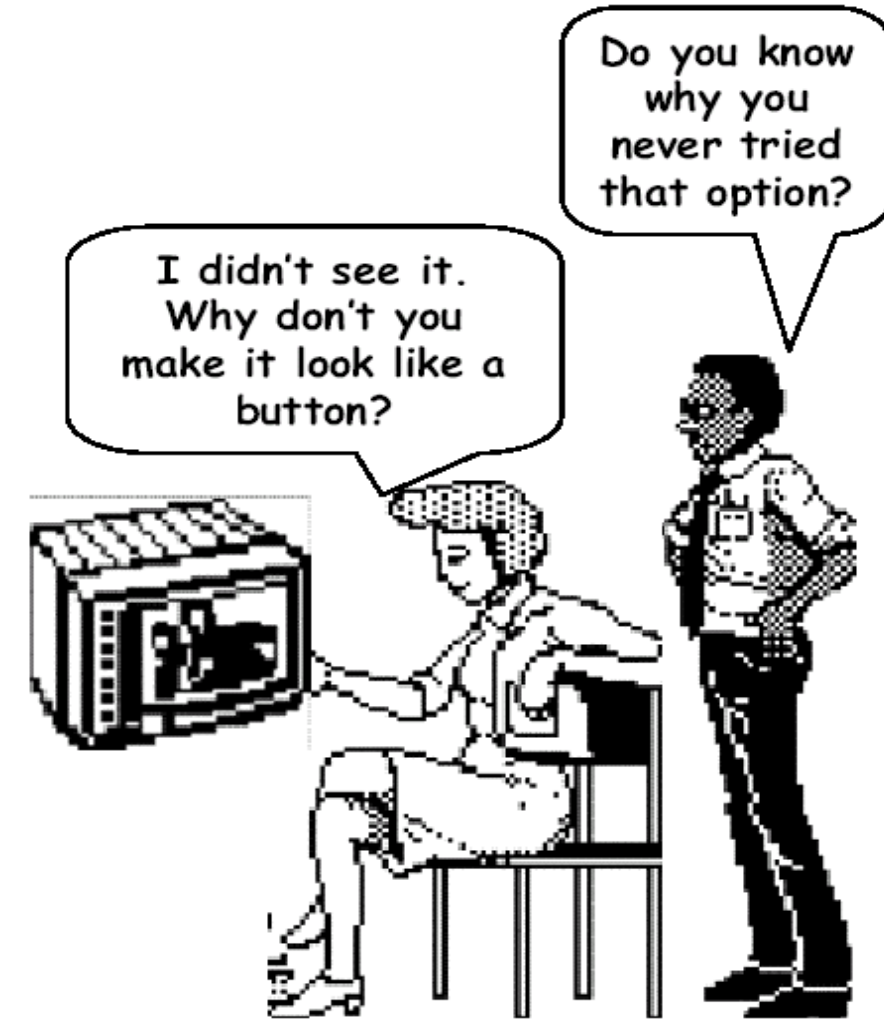
# Retrospective Testing

Additional activity **after an observation**

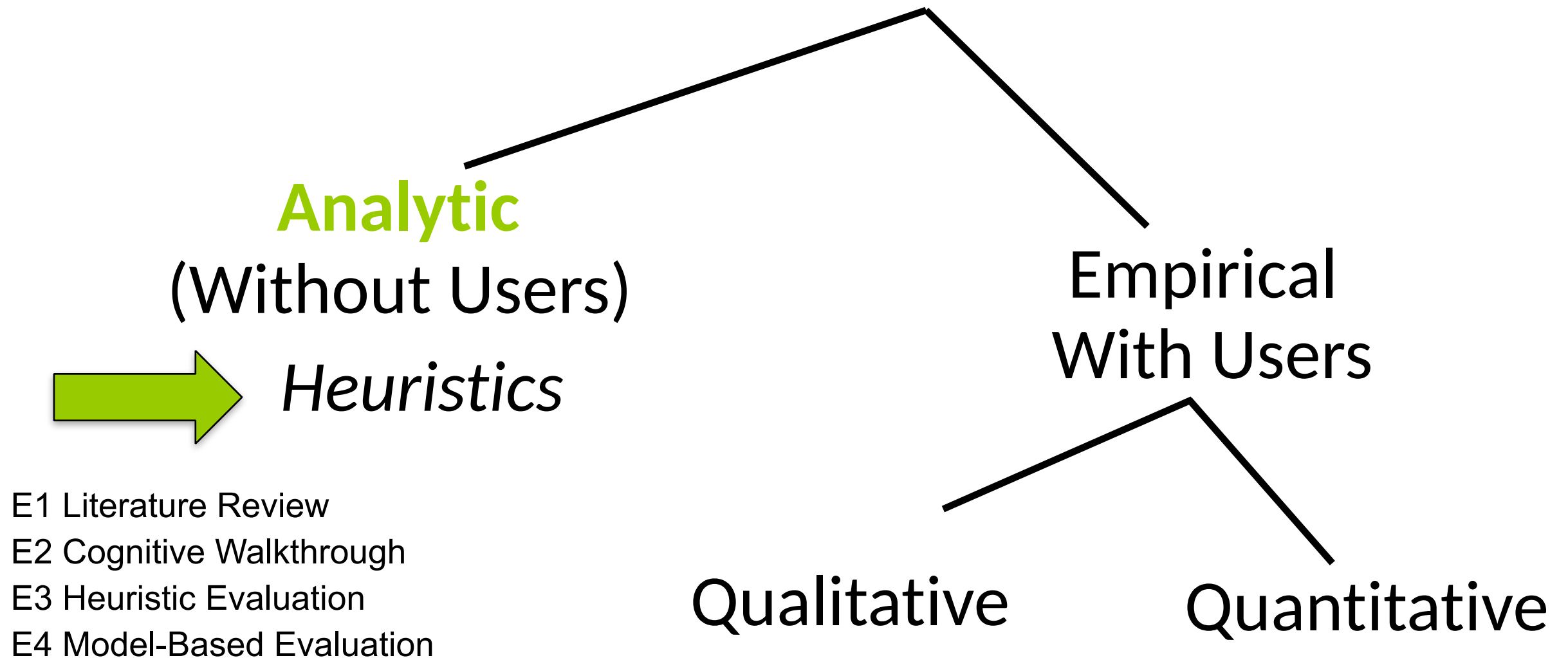
Subject and evaluator **look at video recordings** together,  
user comments his actions retrospectively

Good starting point for **subsequent interview**, looking at video  
avoids wrong memories

Often results in concrete suggestions for improvement



# Evaluation Methods



# Top five plausible excuses for not testing

We don't have the time

We don't have the money

We don't have the expertise

We don't have a usability lab

We wouldn't know how to interpret the results

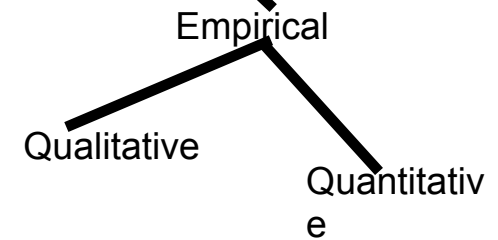
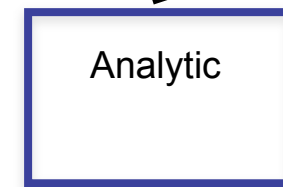
# Heuristic Evaluation

1. recruit a small set (3-5) of “evaluators”
2. evaluators independently check for compliance with usability principles (“heuristics”)
3. different evaluators will find different problems
4. evaluators **only** communicate afterwards
5. findings are then aggregated

  → use list of problems  
to redesign/fix application

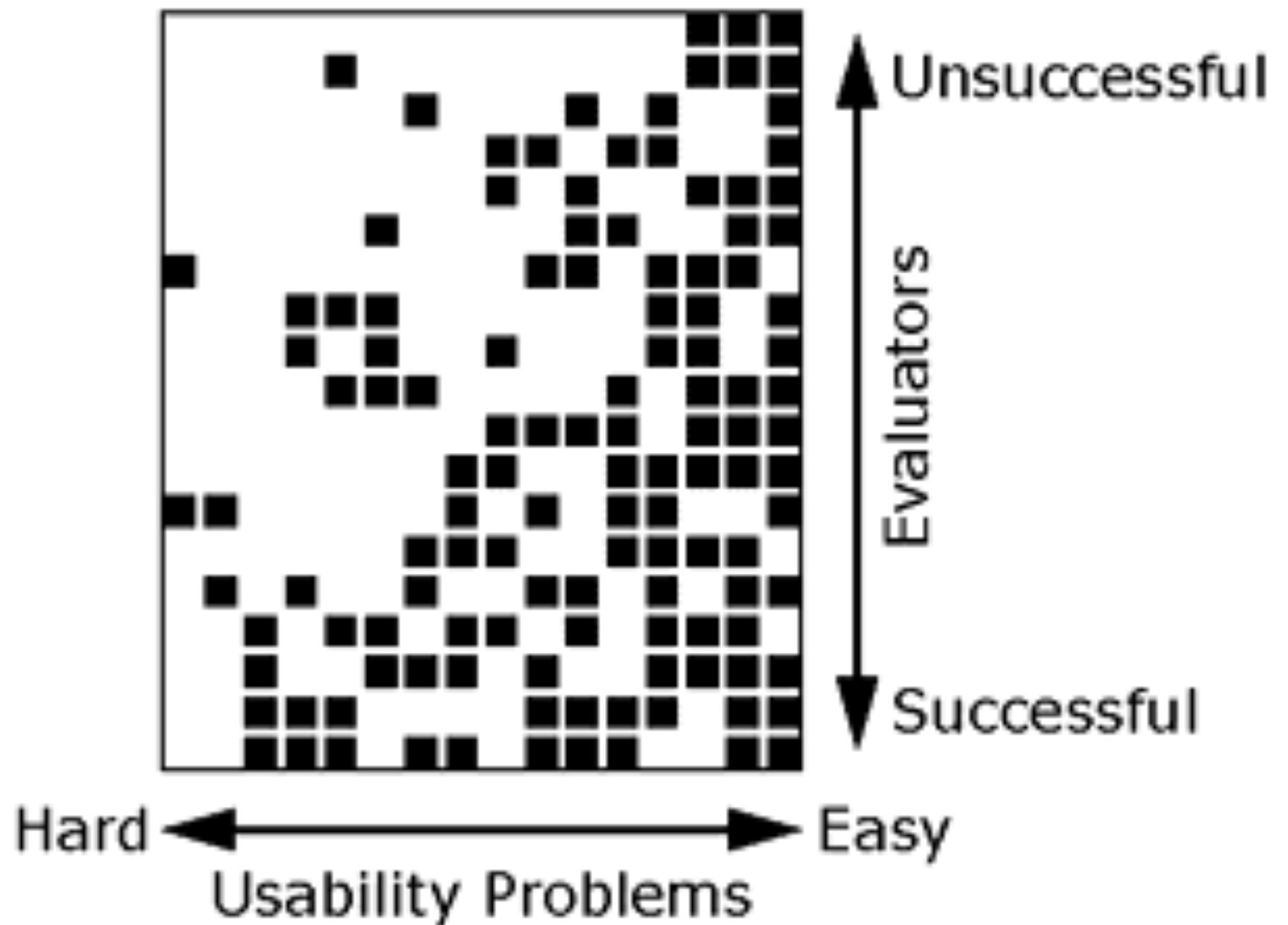


# Heuristic Evaluation

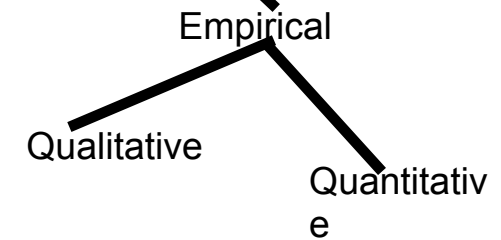
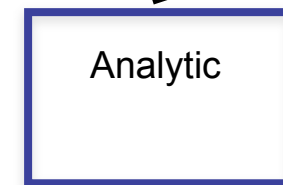


every evaluator does not find every problem

good evaluators find both easy & hard ones

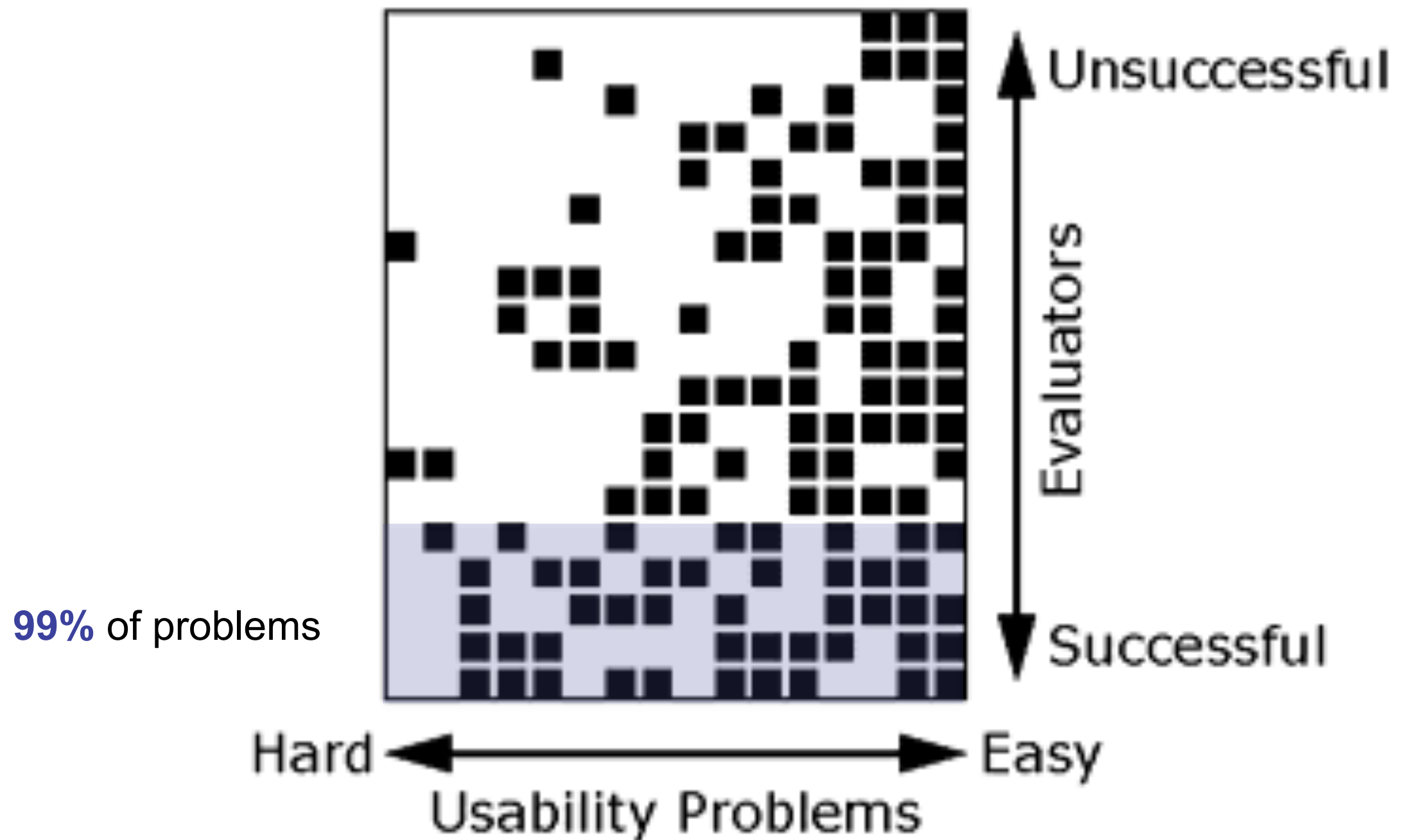


# Heuristic Evaluation

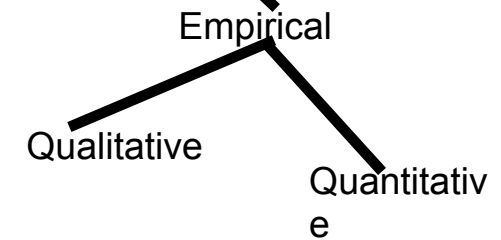
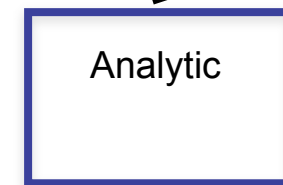


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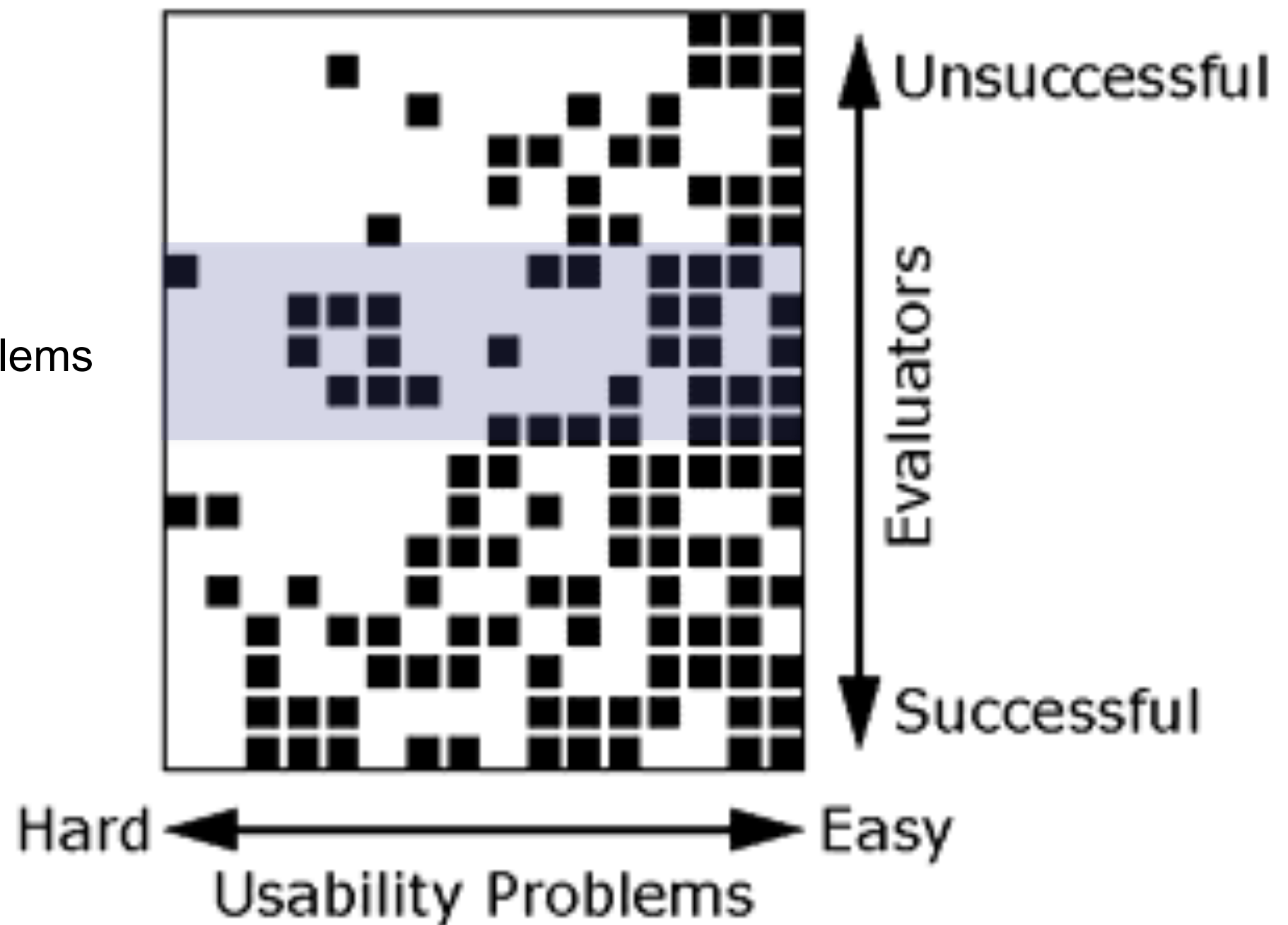


# Heuristic Evaluation

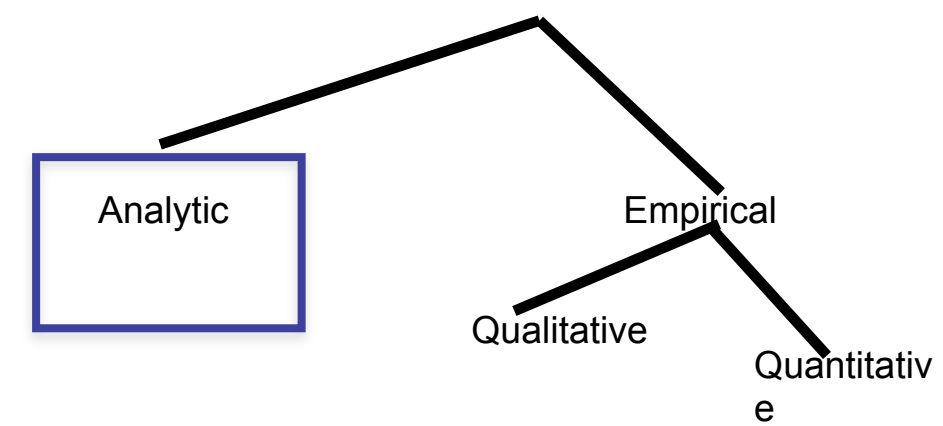


every evaluator does not find every problem  
good evaluators find both easy & hard ones

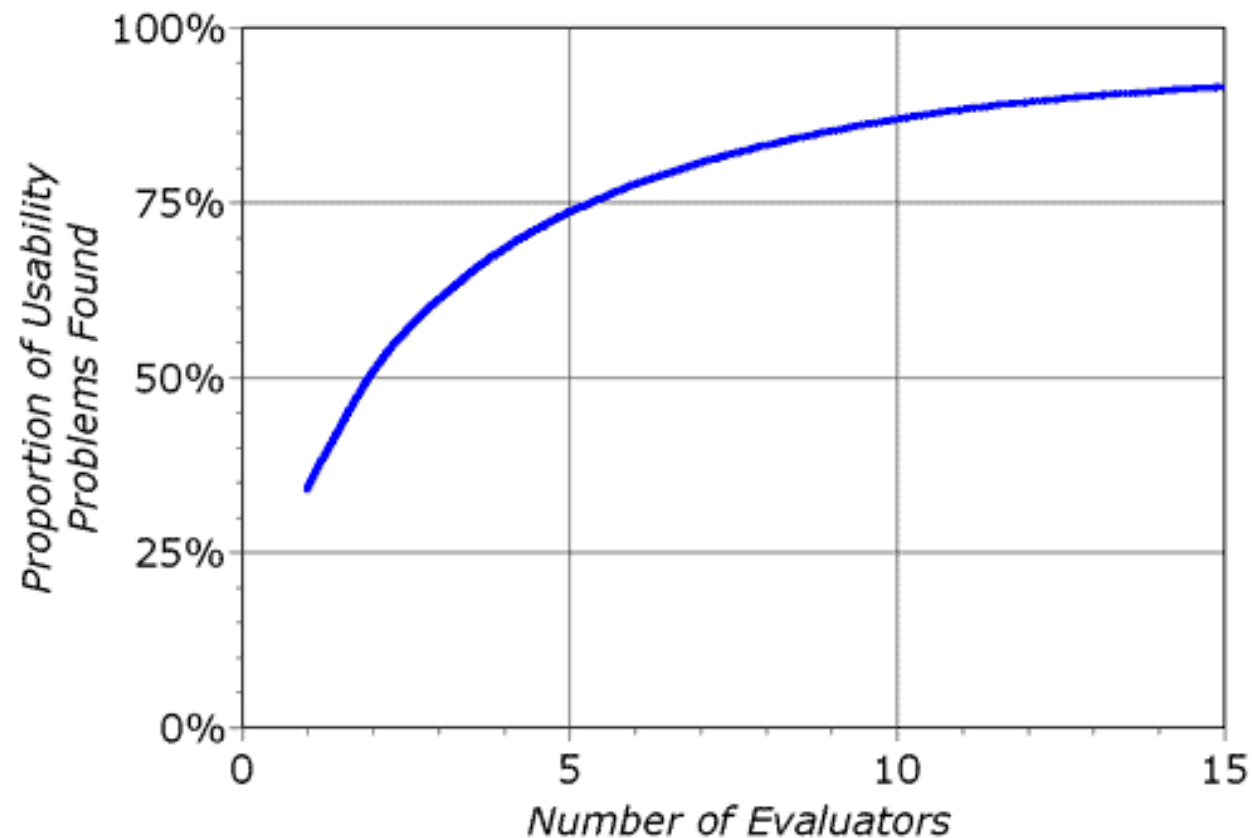
75% of problems



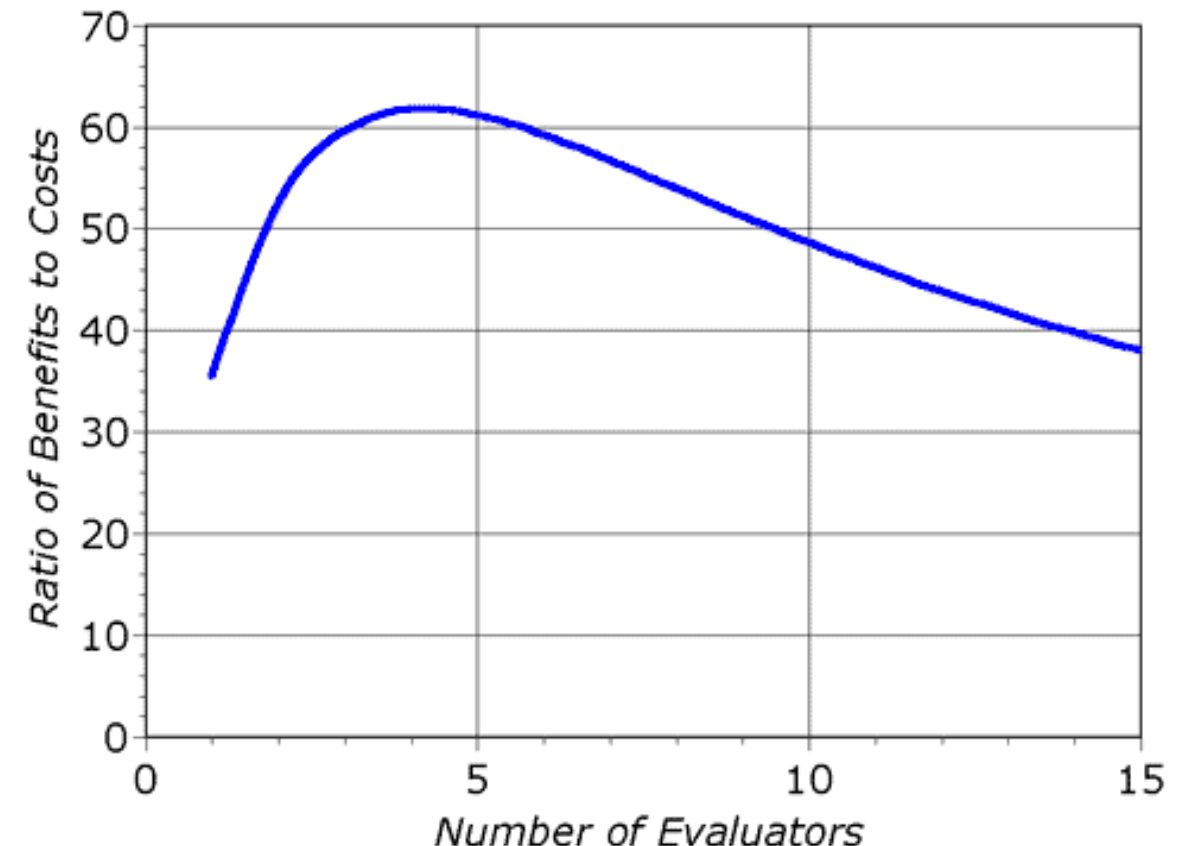
# Heuristic Evaluation



**problems found**



**benefits / cost**



- + Quick and cheap
- Subjective (better done by several independent designers)

See also: [www.useit.com/papers/heuristic](http://www.useit.com/papers/heuristic)

# The heuristics

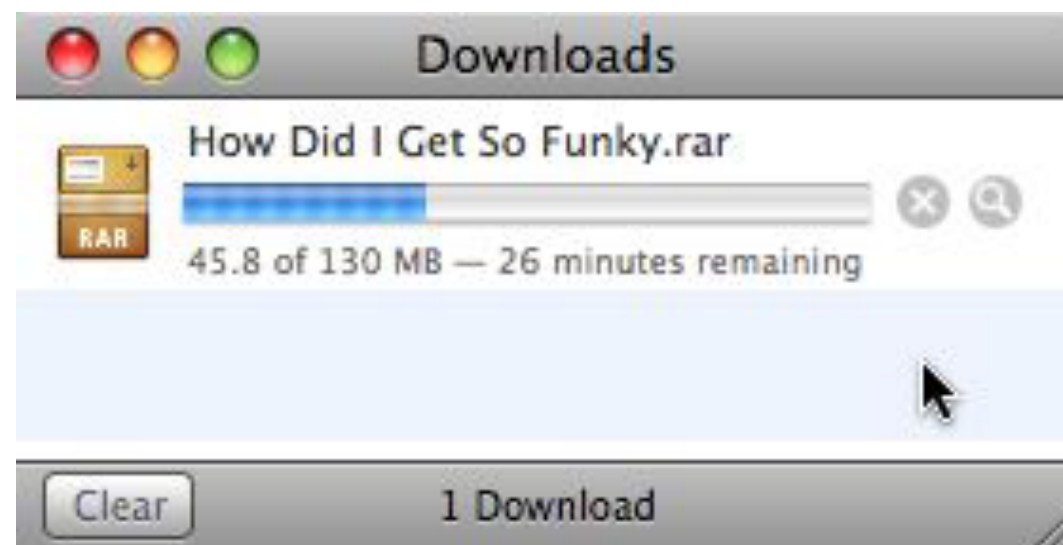
- H2-1 : visibility of system status
- H2-2 : match between system & real world (speak the users' language)
- H2-3 : user control and freedom
- H2-4 : consistency and standards
- H2-5 : error prevention (minimize users' memory load)
- H2-6 : recognition rather than recall
- H2-7 : flexibility and efficiency of use (shortcuts)
- H2-8 : aesthetic & minimalist design
- H2-9 : help recognize, diagnose, & recover from errors
- H2-10 : help and documentation



# visibility of system status

pay attention to response time

- 0.1 sec: no special indicators
- 1.0 sec: user tends to lose track of data
- 10 sec: max. duration if user to stay focused on action
- for longer delays, use percent-done progress bars





**100ms** delay results in 1% sales loss.  
(potential \$191M in lost revenue in 2008)

---



**400ms** delay results in 5-9% drop in full-page traffic.

---



**500ms** delay drops search traffic by 20%.  
The cost of slower performance increases over time.

---



**1s** delay results 4% drop in revenue

---



Fastest 10% of users stay 50% longer than slowest 10%

---



**Your password has been emailed.**

**Theresa Neil sign in**

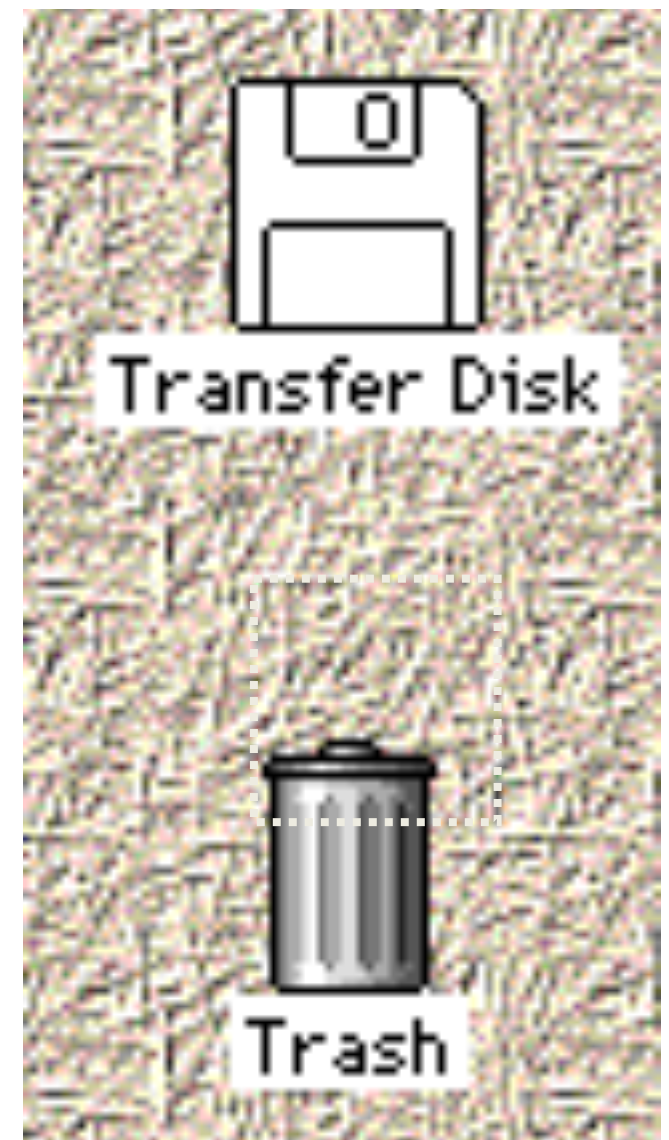
---

# match between system & real world

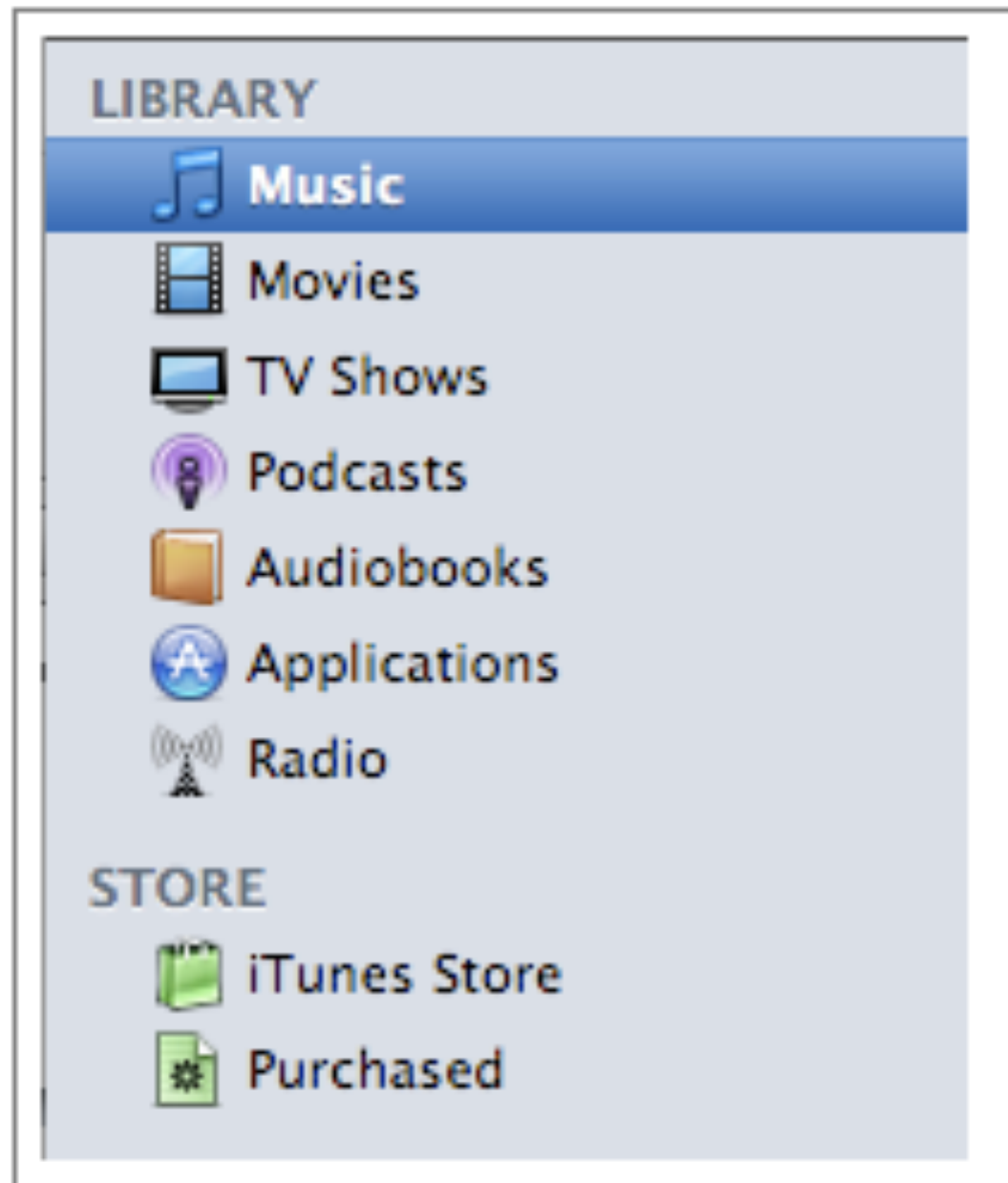
speak the users' language  
follow real world conventions

**example of violation:**  
dragging disk to Mac trash  
**should delete it**, not eject it

**bad**



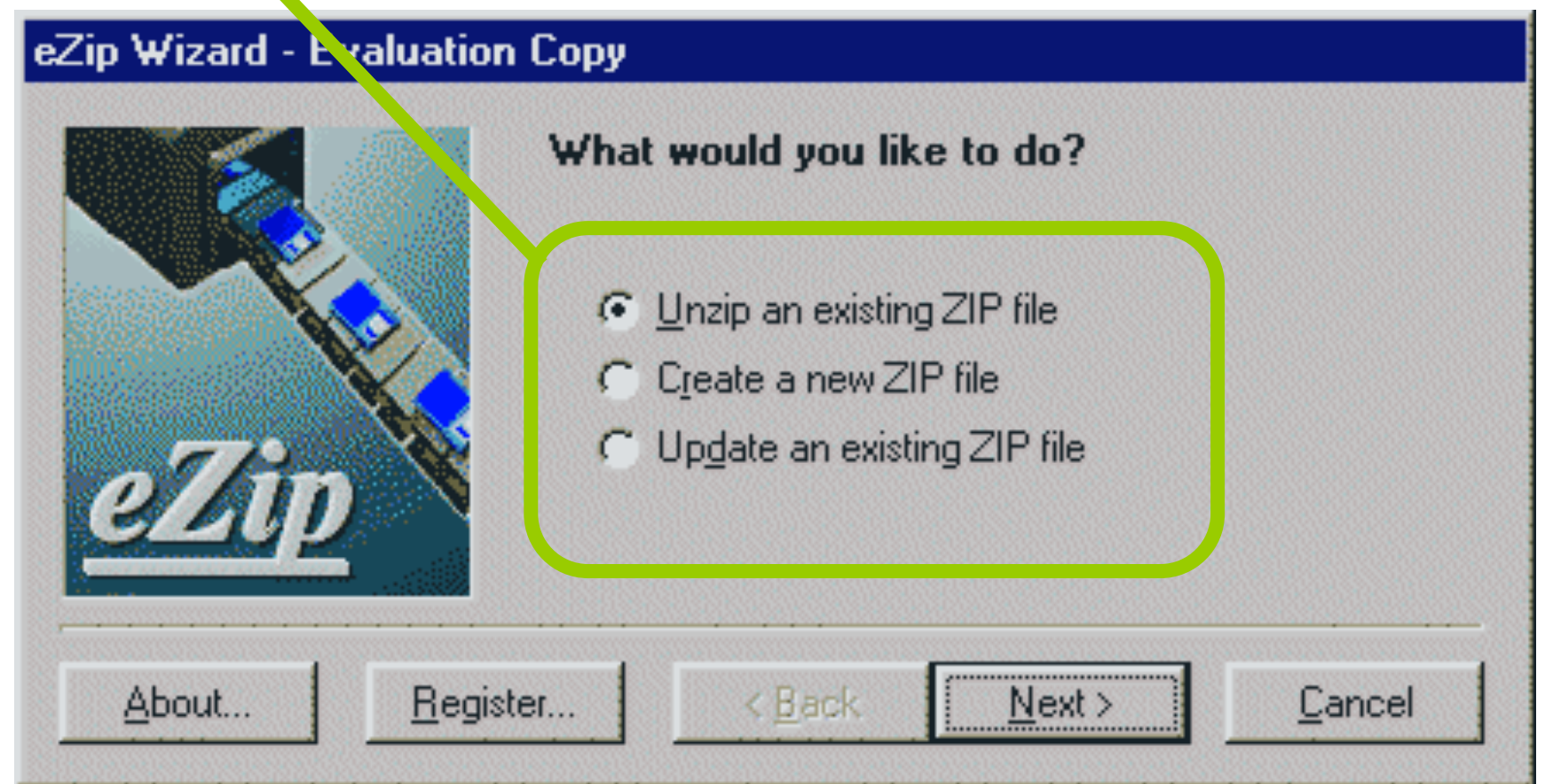
# match between system & real world



# user control & freedom

–offer “exits” for mistaken choices, undo, redo

good



- **wizards:** must respond to question before going to next
- good for infrequent tasks (e.g., modem config.) and beginners
- not for common tasks and experts → **have 2 versions (WinZip)**

The Wufoo  
**FORM**  
GALLERY

Search

Forms

Surveys

Invitations

Registrations

Lead Generation

Online Orders

Survey  
TEMPLATES

1 Customer Satisfaction Survey

2 Cancellation Survey

3 Business Demographic Survey

4 Web Site Visitor Survey

5 Tech Support Satisfaction Survey

6 Health Survey

Download HTML

Add to Wufoo

Wufoo

## Customer Satisfaction Survey

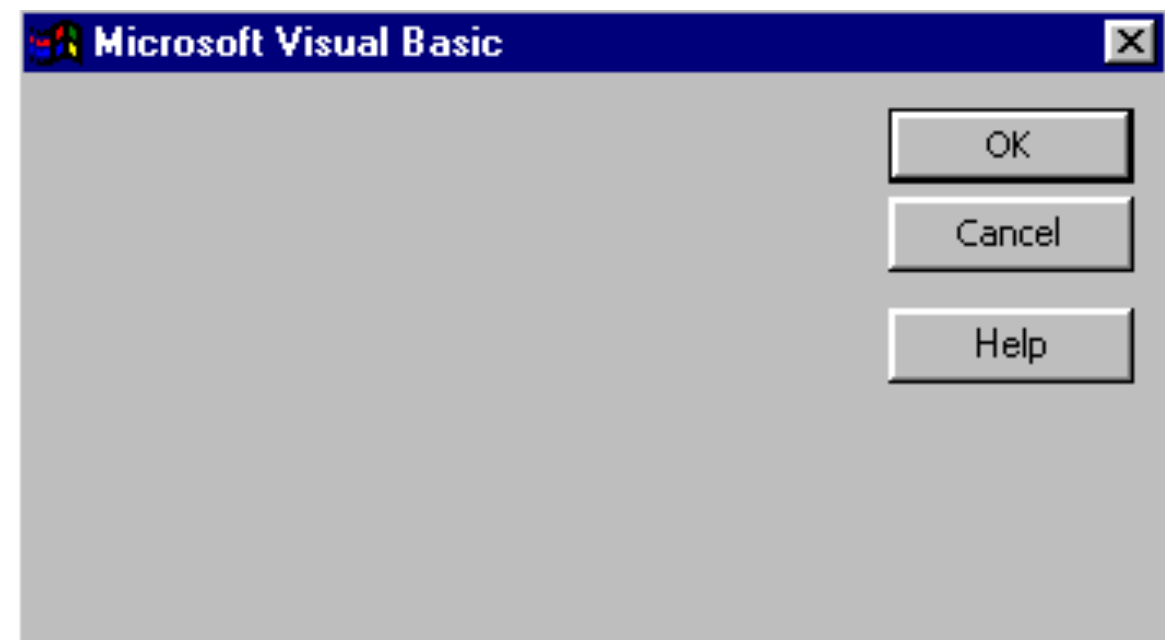
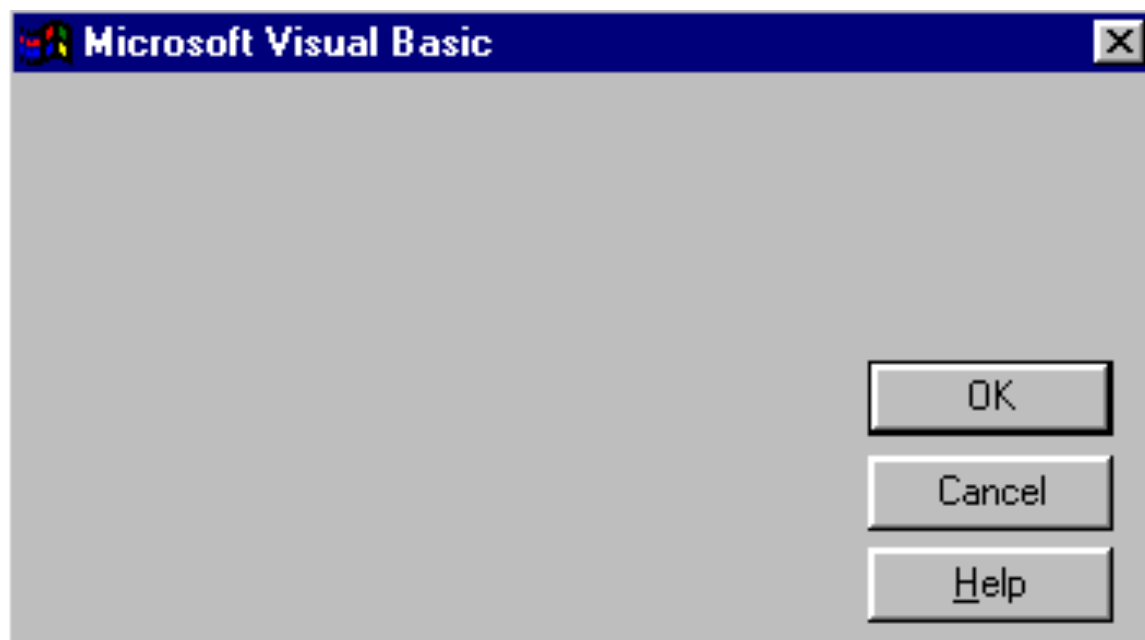
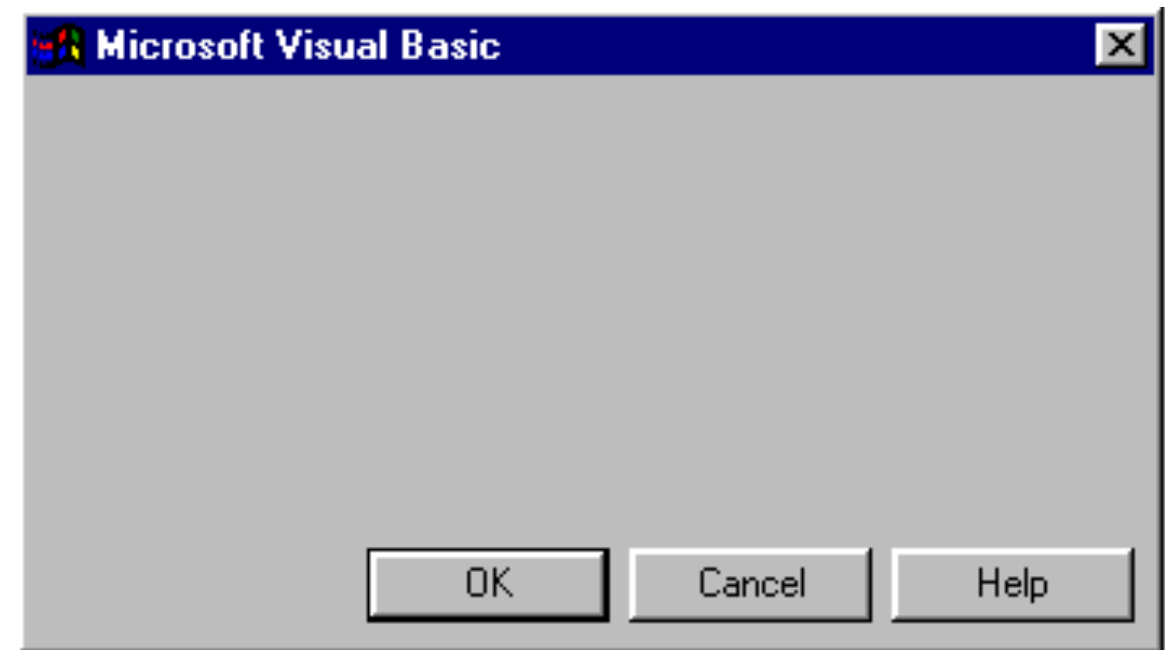
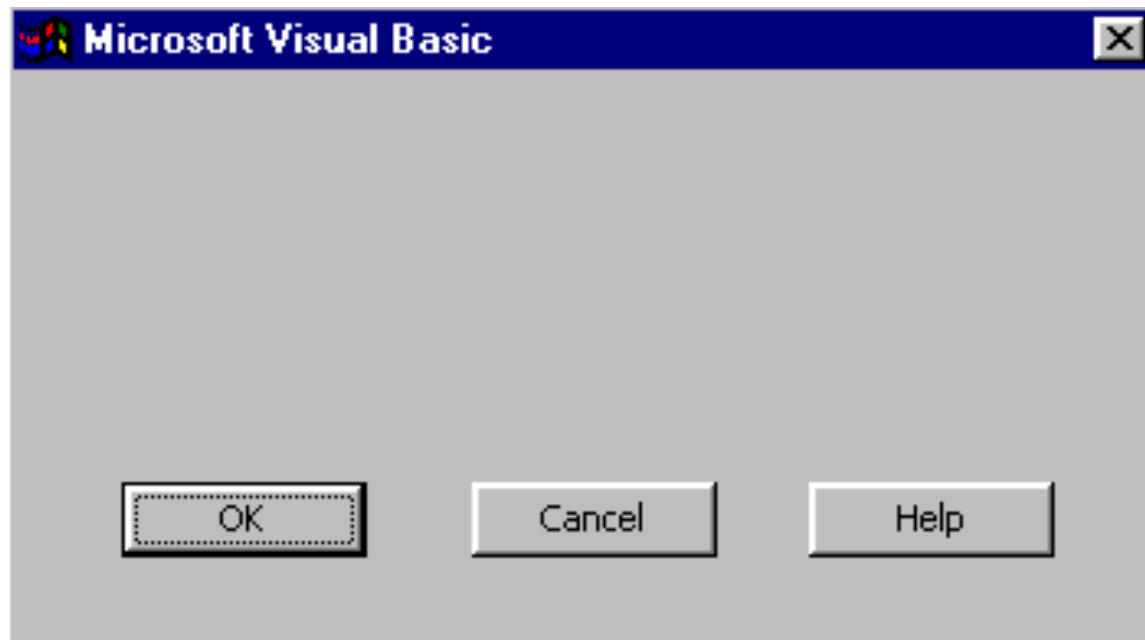
Please take a few moments to complete this satisfaction su

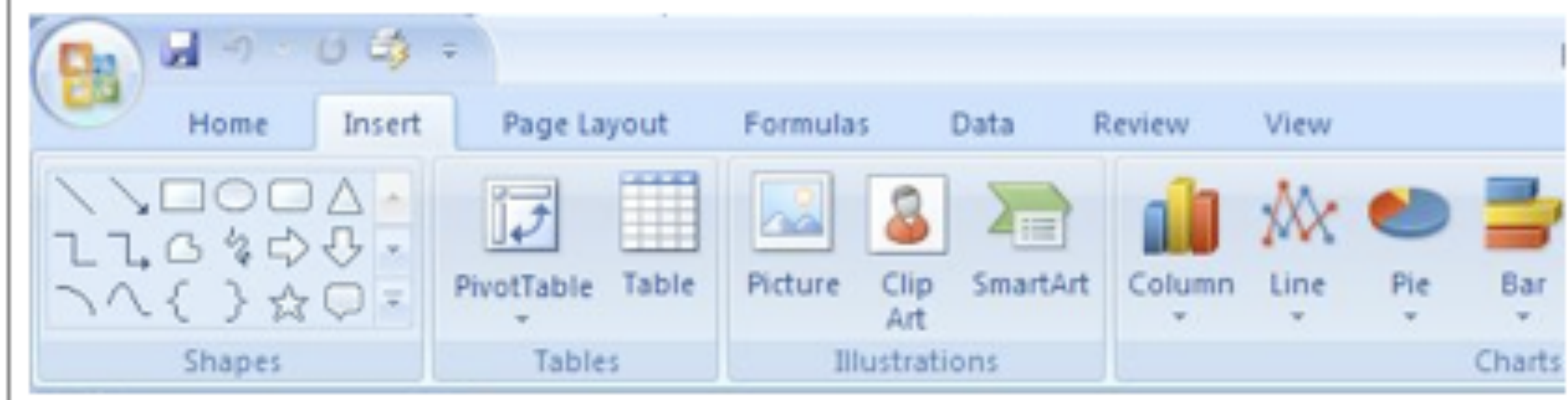
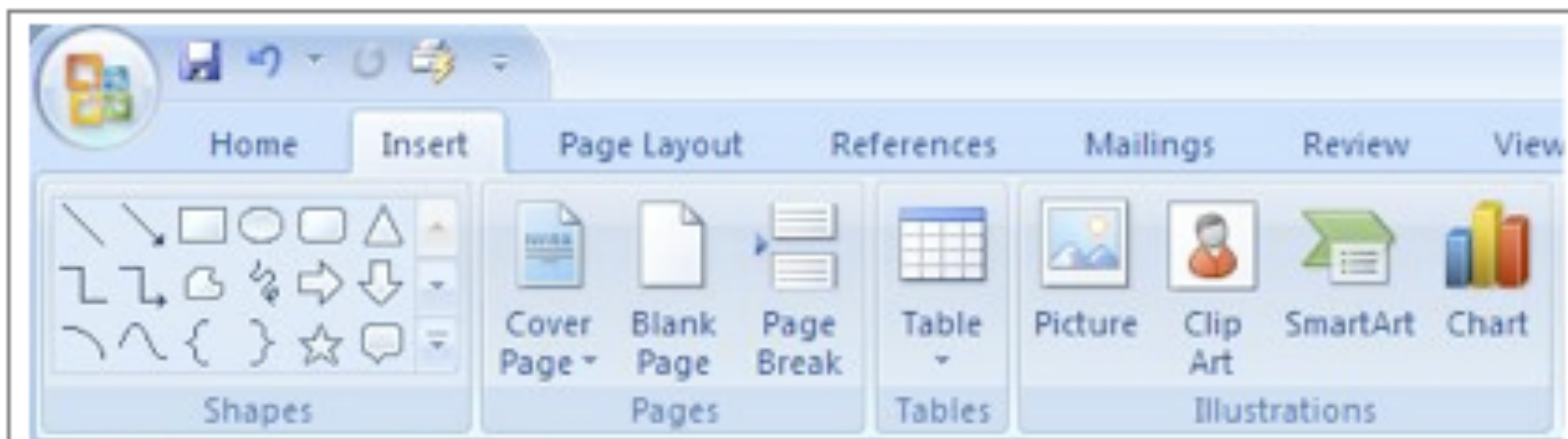
How long have you used our product / service?

- ☐ Less than a month
- ☐ 1-6 months
- ☐ 1-3 years
- ☐ Over 3 Years

# Consistency & standards

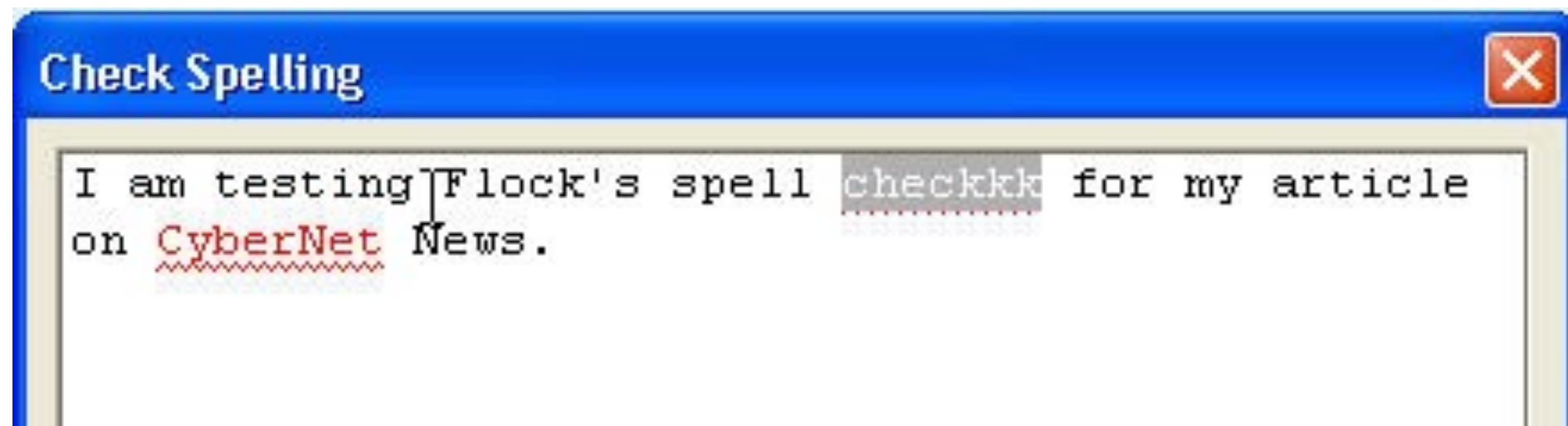
bad





# error prevention

good



# error prevention

Share something with **Usabilitypost:**

Update

 [Attach file](#)

# error prevention

|                     |                     |                                 |
|---------------------|---------------------|---------------------------------|
| design              |                     | <a href="#">Advanced Search</a> |
|                     |                     | <a href="#">Preferences</a>     |
|                     |                     | <a href="#">Language Tools</a>  |
| design within reach | 5,350,000 results   |                                 |
| designer handbags   | 3,430,000 results   |                                 |
| designer shoes      | 2,630,000 results   |                                 |
| designer clothes    | 3,120,000 results   |                                 |
| designer dresses    | 1,110,000 results   |                                 |
| design sponge       | 9,930,000 results   |                                 |
| designer            | 285,000,000 results |                                 |
| design museum       | 13,600,000 results  |                                 |
| designers guild     | 530,000 results     |                                 |
| designer jeans      | 2,010,000 results   |                                 |
|                     |                     | <a href="#">close</a>           |

# Recognition rather recall



Please select your country or territory

Please select your country or territory

United States  
United Kingdom  
Albania  
Algeria  
American Samoa  
Andorra  
Angola  
Anguilla  
Antarctica  
Antigua and Barbuda  
Argentina  
Armenia  
Aruba  
Australia  
Austria  
Azerbaijan  
Bahamas  
Bahrain  
Bangladesh

Arno Pro

Ayuthaya

Baghdad

✓ **BANK GOTHIC**

Baskerville

Baskerville Old Face

Bastion

Batang

# flexibility and efficiency of use

- accelerators for experts (e.g., gestures, kb shortcuts)
- allow users to tailor frequent actions (e.g., macros)

## Google Docs Keyboard Shortcuts dashkards http://dashkards.com/

| File                   |     | Editing        |     |
|------------------------|-----|----------------|-----|
| Open...                | ⌘O  | Insert Comment | ⌘⌘M |
| Print                  | ⌘P  | Insert Link    | ⌘K  |
| Save                   | ⌘S  | Find           | ⌘F  |
| Formatting             |     | Select All     | ⌘A  |
| Bold                   | ⌘B  | Copy           | ⌘C  |
| Italic                 | ⌘I  | Cut            | ⌘X  |
| Underline              | ⌘U  | Paste          | ⌘V  |
| Header Style n         | ⌘⇧n | Undo           | ⌘Z  |
| n=1...6                |     | Redo           | ⌘Y  |
| Normal Paragraph Style | ⌘⇧0 | Define...      | ⌘⇧Y |
| Toggle Numbered List   | ⌘⇧7 | Word Count     | ⌘⇧C |
| Toggle Bulleted List   | ⌘⇧8 |                |     |
| Align Left             | ⌘⇧L |                |     |
| Align Centered         | ⌘⇧E |                |     |
| Align Right            | ⌘⇧R |                |     |
| Justify                | ⌘⇧J |                |     |
| Clear Formatting       | ⌘\  |                |     |
| Superscript            | ⌘.  |                |     |
| Subscript              | ⌘,  |                |     |

⌘ Command

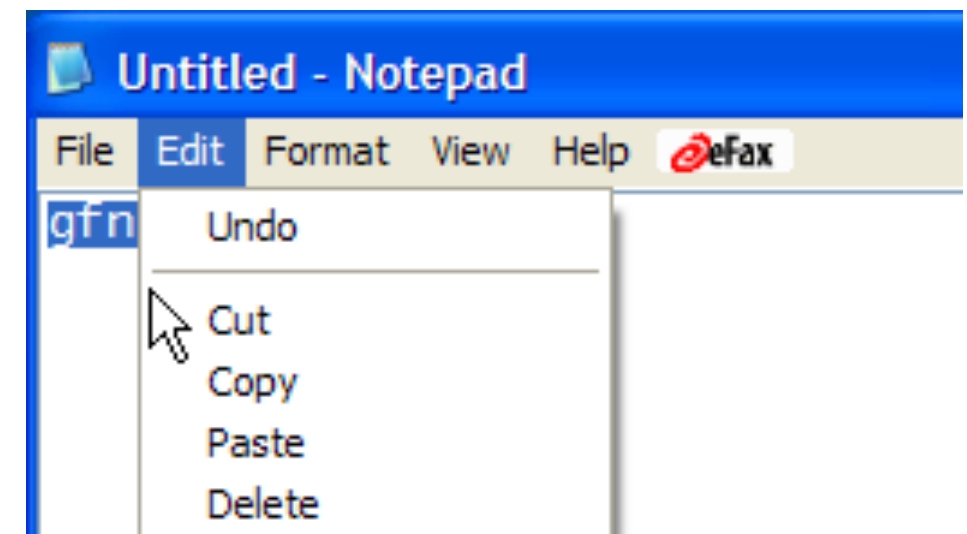
⇧ Shift

⇥ Tab

⌥ Option

^ Control

↵ Enter

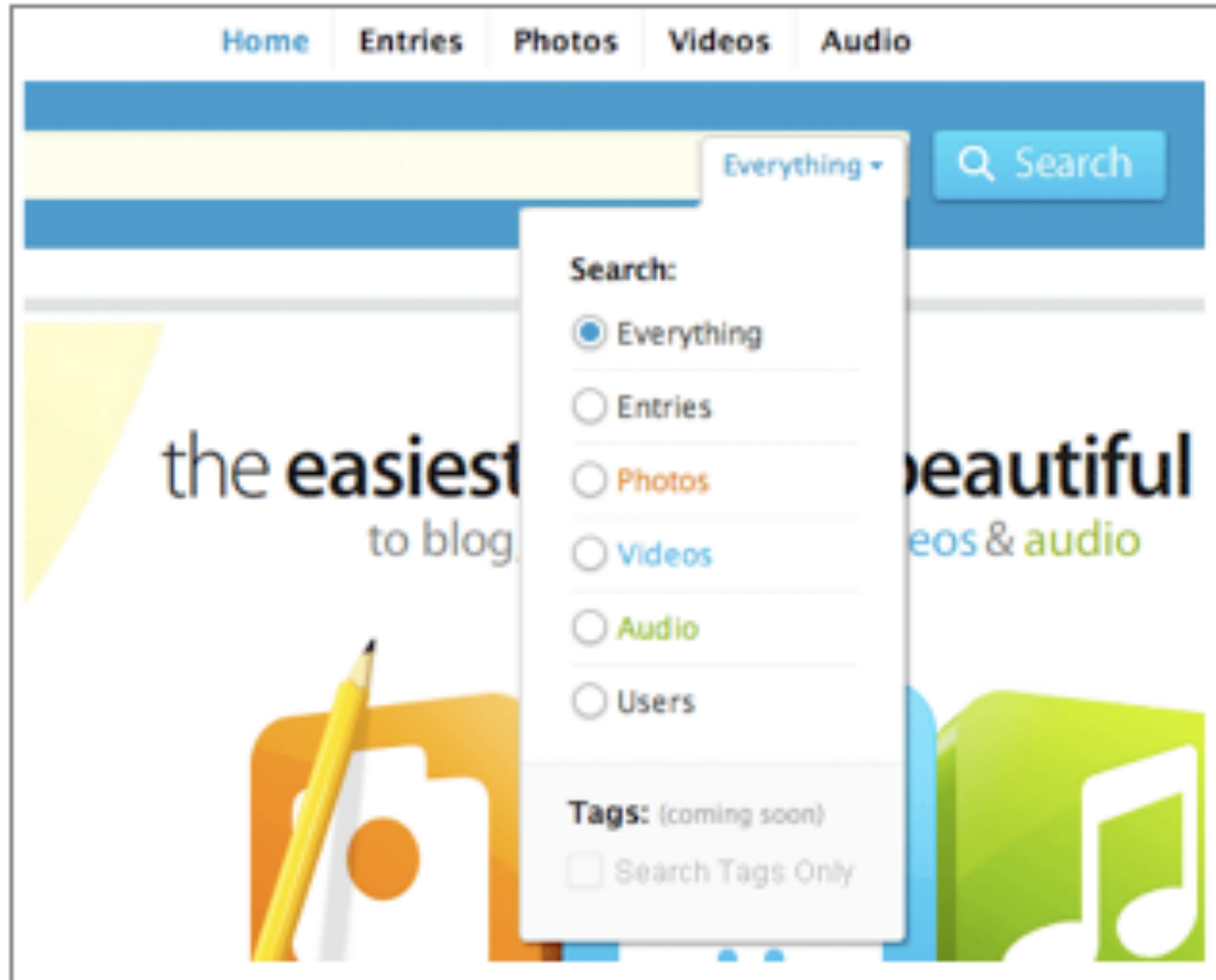


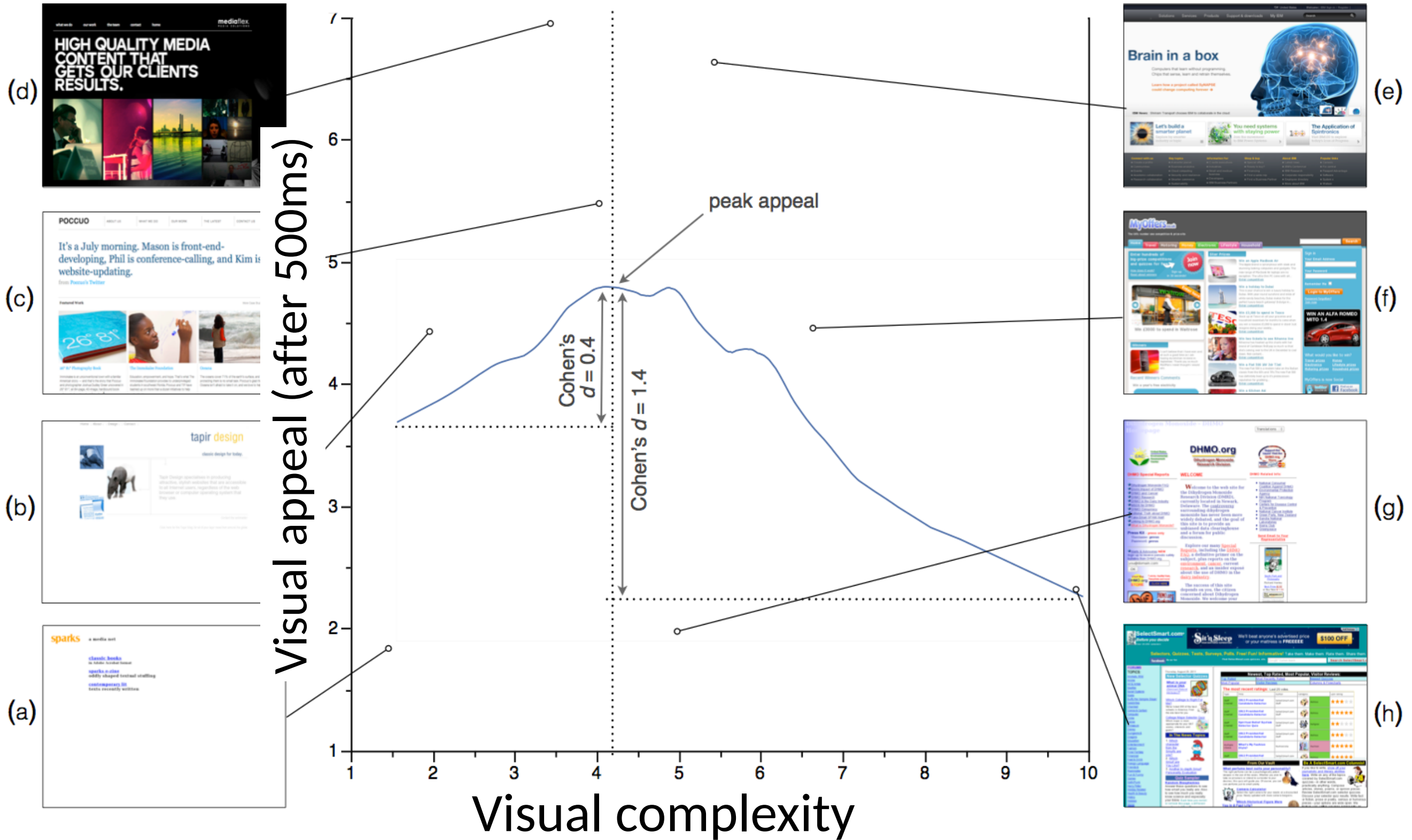
# aesthetic & minimalist design

- avoid irrelevant information in dialogues

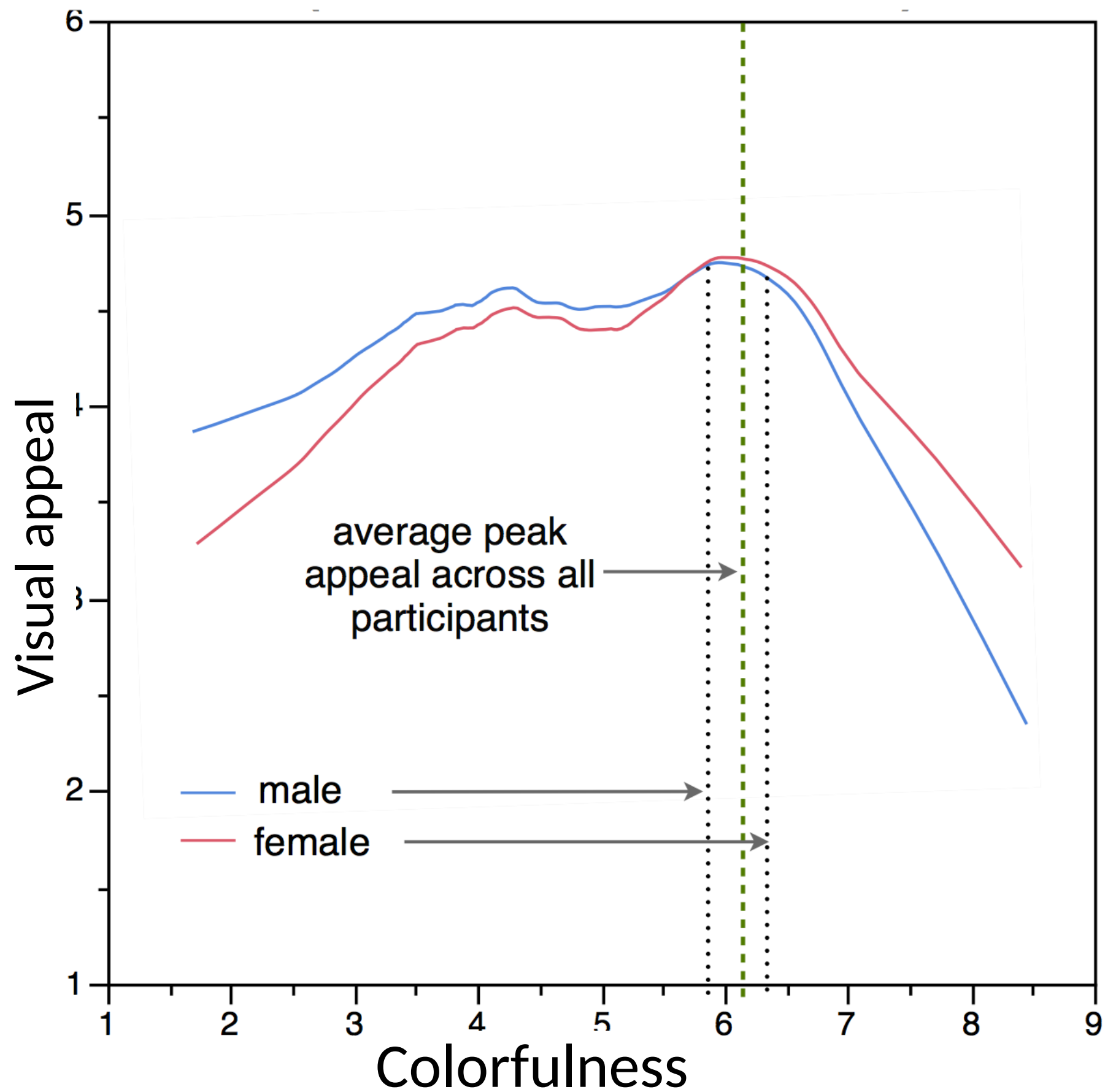
|                                                                                |                                   |                                      |
|--------------------------------------------------------------------------------|-----------------------------------|--------------------------------------|
| Form Title -- (appears above URL in most browsers and is used by W/W/W search) |                                   | Background Color:                    |
| Q&D Software Development Order Desk                                            |                                   | FFFBF0                               |
| Form Heading -- (appears at top of Web page in bold type)                      |                                   | Text Color:                          |
| Q&D Software Development Order Desk <input checked="" type="checkbox"/> Center |                                   | 000080                               |
| E-Mail responses to (will not appear on)                                       | Alternate (for mailto forms only) | Background Graphic                   |
| dversch@q-d.com                                                                |                                   |                                      |
| Text to appear in Submit button                                                | Text to appear in Reset button    | <input type="radio"/> Mailto         |
| Send Order                                                                     | Clear Form                        | <input checked="" type="radio"/> CGI |
| Scrolling Status Bar Message (max length = 200 characters)                     |                                   |                                      |
| ***WebMania 1.5b with Image Map Wizard is here!***                             |                                   |                                      |
| << Prev Tab                                                                    |                                   | Next Tab >>                          |

# Aesthetic & Minimalist design





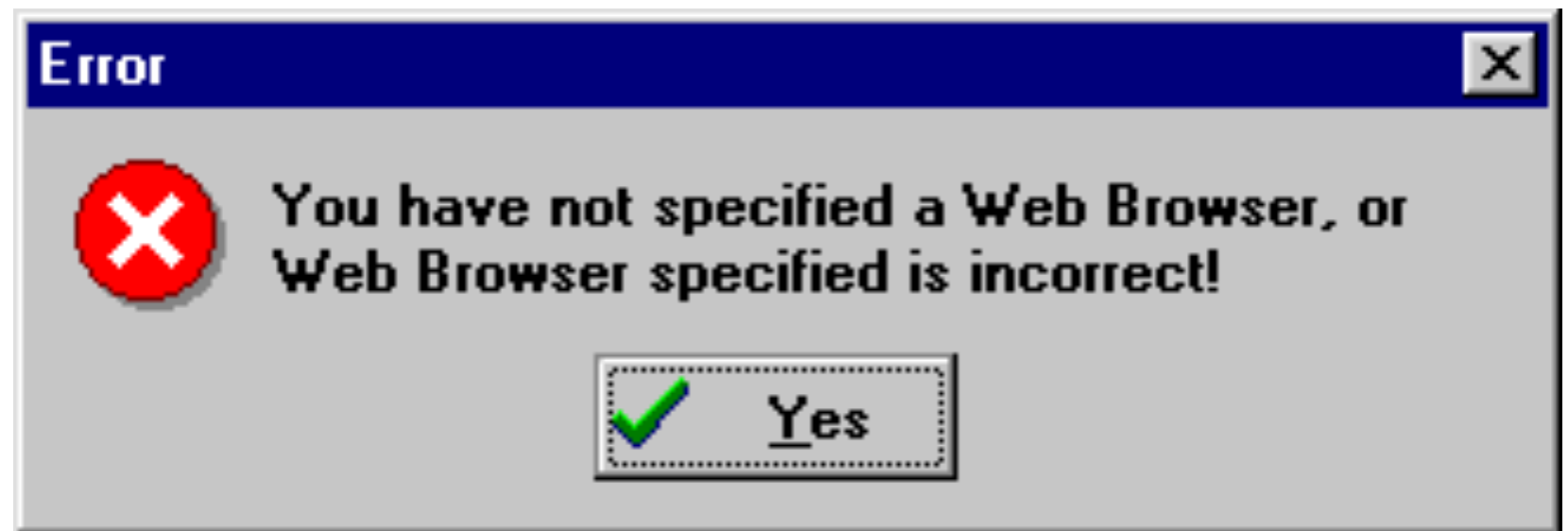
Katharina Reinecke and Krzysztof Z. Gajos. 2014. Quantifying visual preferences around the world. ACM CHI '14 11-20.



(b) Gender differences

# help recognize, diagnose, & recover from errors

- error messages in plain language
- precisely indicate the problem
- constructively suggest a solution



# help & documentation

## Or start a new account

Choose a username (no spaces)

 bert is already taken. Please choose a different username.

Choose a password

 Passwords must be at least 6 characters and can only contain letters and numbers.

Retype password

Email address (must be real!)

 The email provided does not appear to be valid

☒ Send me occasional Digg updates.



# Oh no!

It seems the page you were trying to find on my site isn't around anymore (or at least around here).

[Report it missing using my contact form](#) and I'll see what I can do about it.

Whilst your here why not check out my [articles listing](#) or [browse my blog](#)? You never know - you may just

## **#1 GUIDAGE**

Incitation

Groupement / Distinction

Feedback Immédiat

Lisibilité

## **#2 CHARGE DE TRAVAIL**

Brièveté

Densité de l'information

## **#3 CONTRÔLE DES UTILISATEURS SUR LEURS ACTIONS**

Actions explicites

Contrôle utilisateur

## **#4 ADAPTABILITÉ DE L'APPLICATION**

Flexibilité

Prise en compte de l'expérience

## **#5 GESTION DES ERREURS**

Protection contre les erreurs

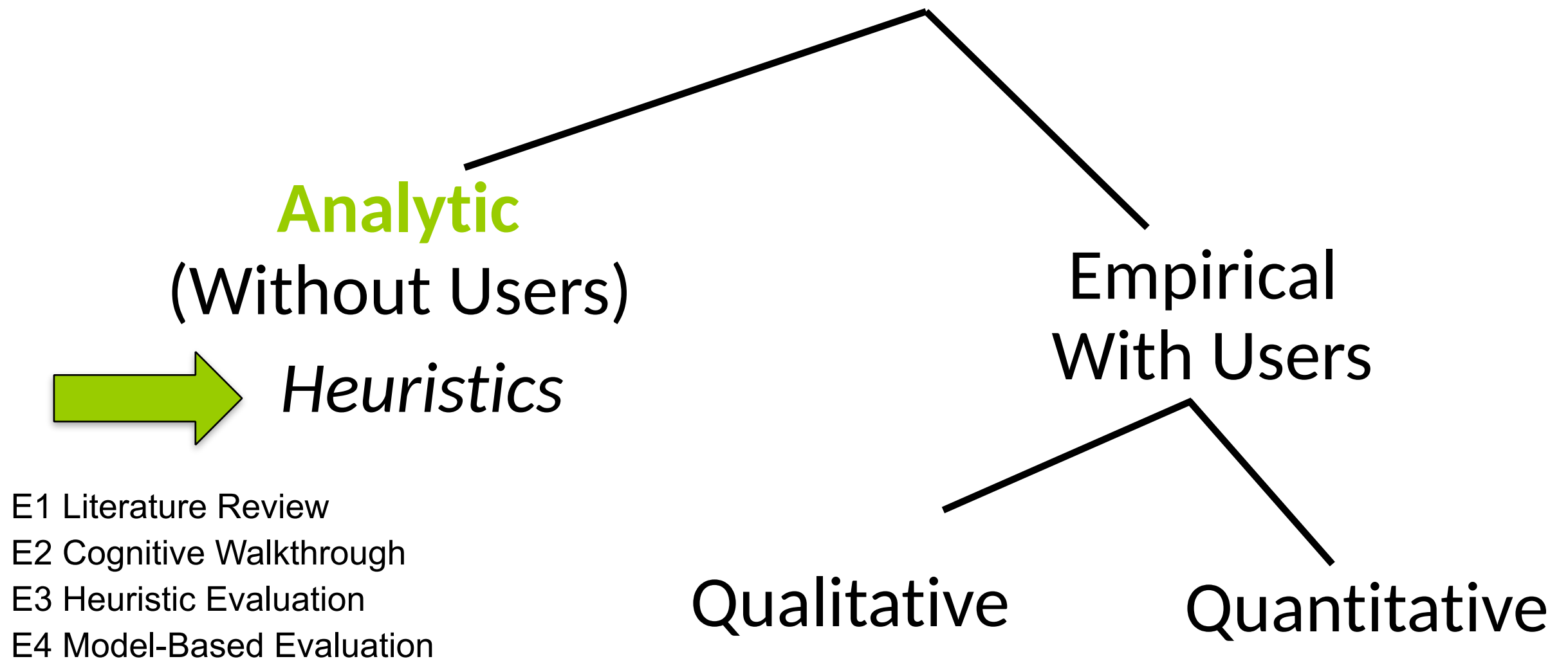
Qualité des messages d'erreur

## **#6 HOMOGÉNÉITÉ ET COHÉRENCE**

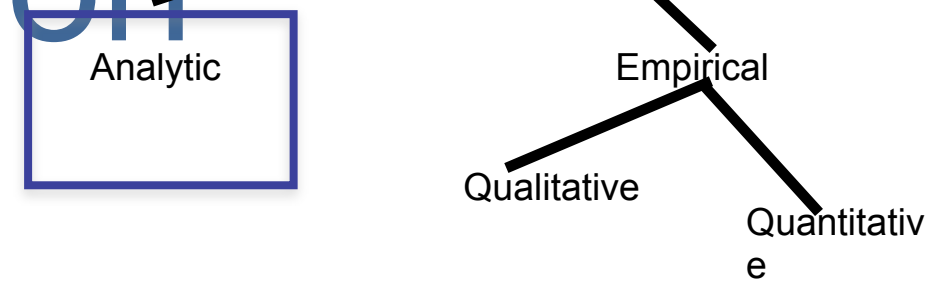
## **#7 SIGNIFIANCE DES CODES ET DÉNOMINATIONS**

## **#8 COMPATIBILITÉ**

# Evaluation Methods



# Model-Based Evaluation



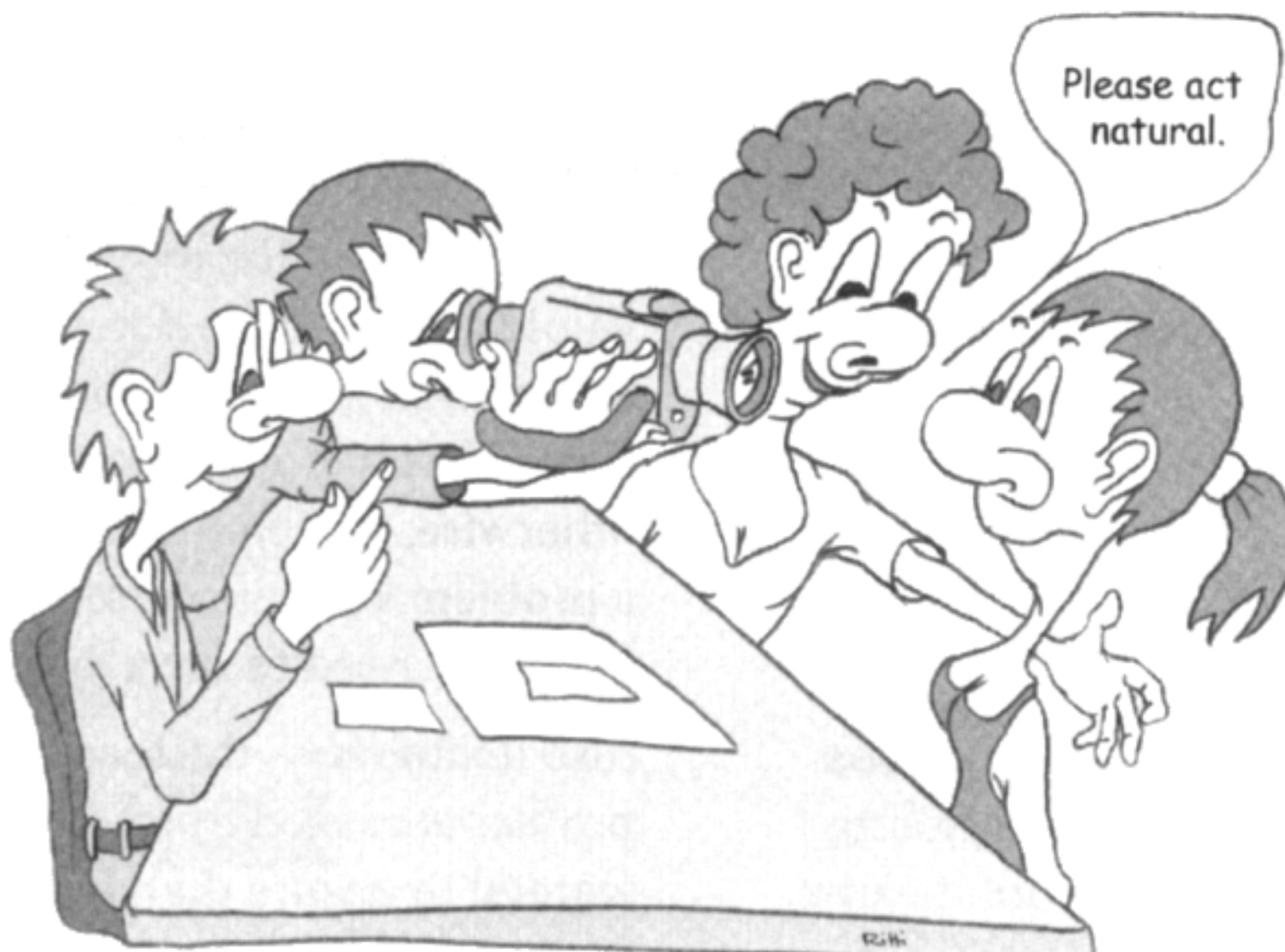
Several theoretical models exist that offer a framework for design and evaluation

## Examples

- **GOMS**, KLM
- Design Rationale (history of design decisions with reasons and alternatives)
- Design Patterns

Next year:

Environment Hautement interactif

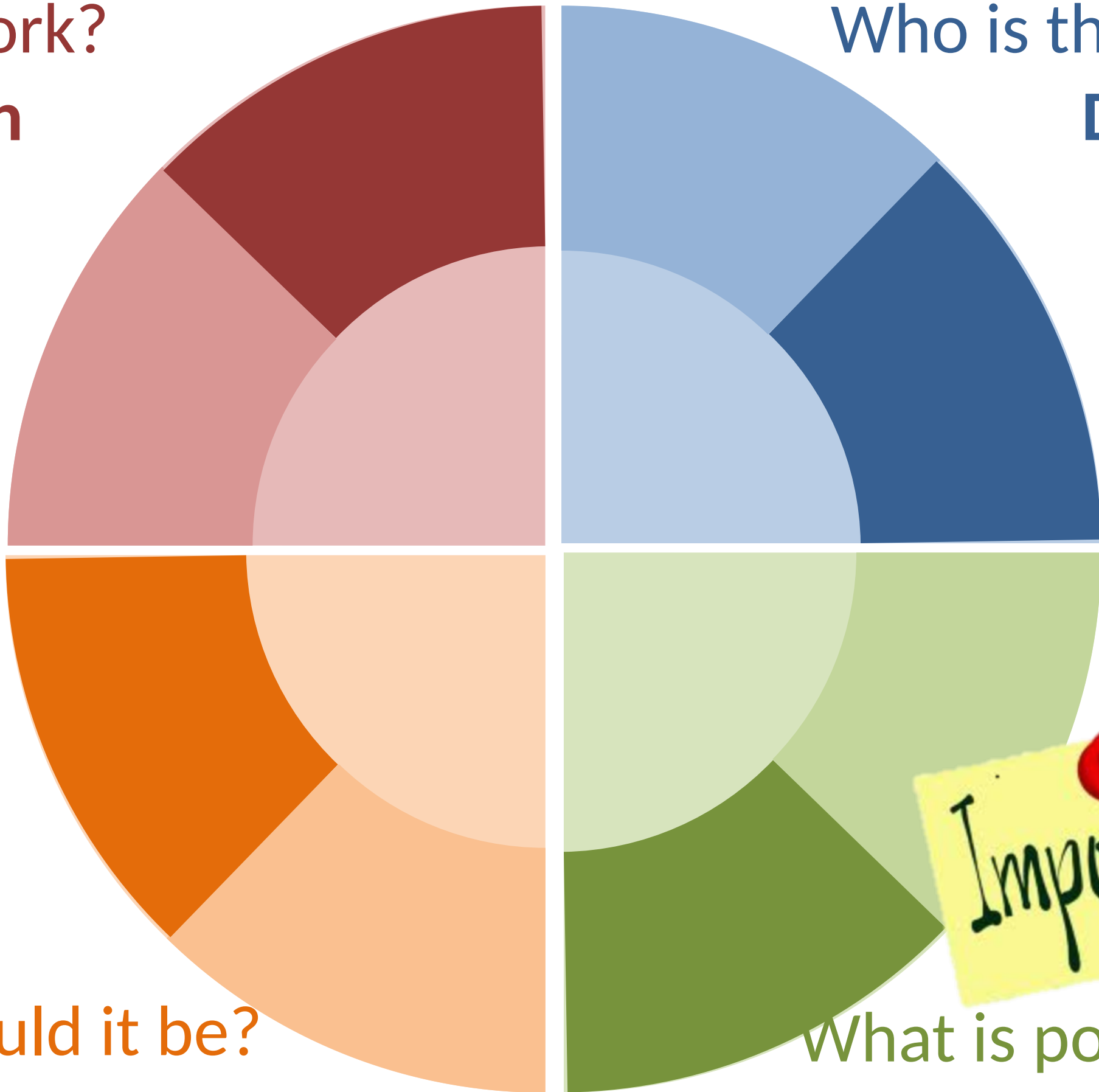


Does it work?  
**Evaluation**

Who is the user?  
**Discover**

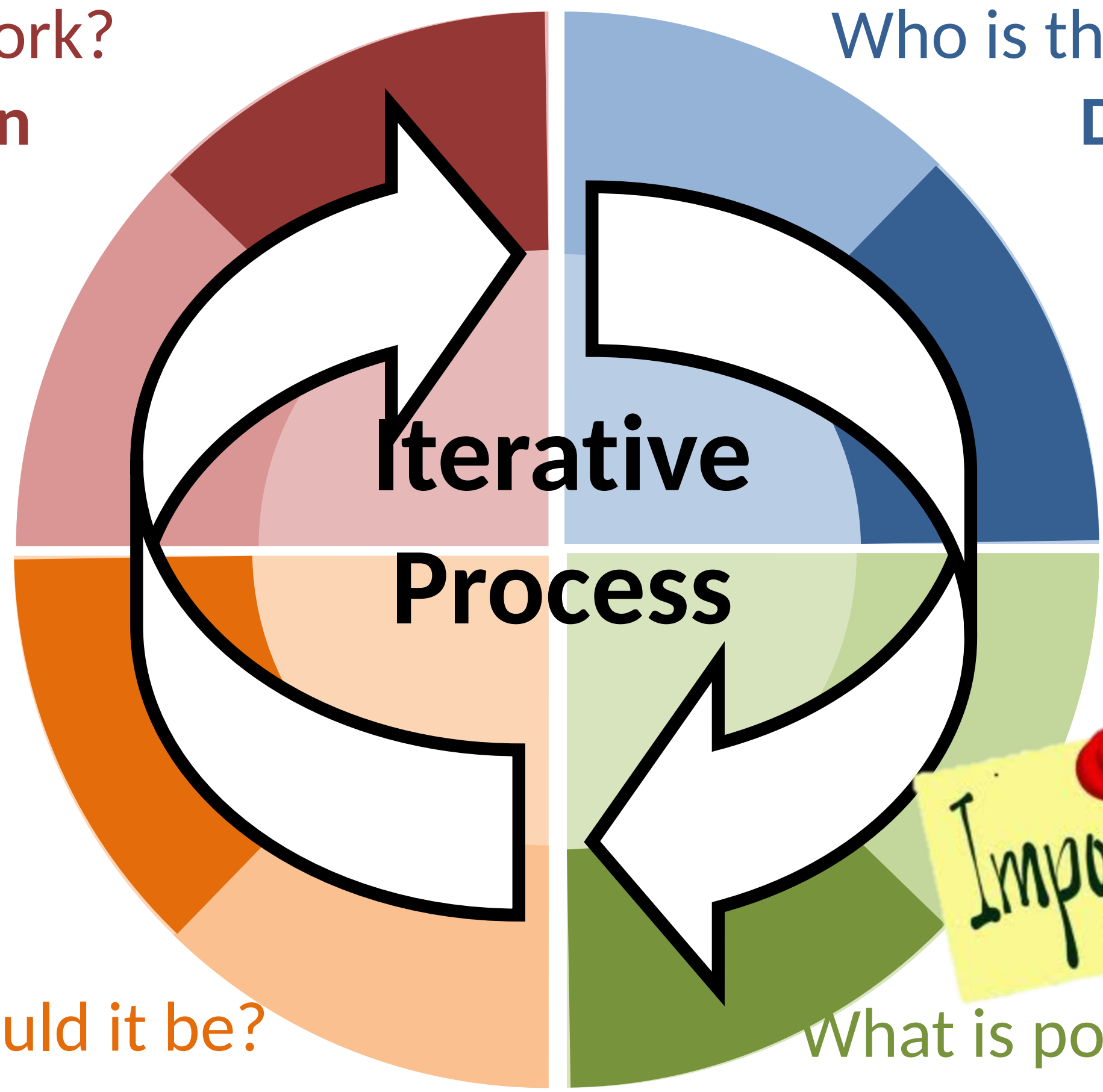
What should it be?  
**Design**

What is possible?  
**Invention**



Does it work?  
**Evaluation**

Who is the user?  
**Discover**



What should it be?  
**Design**

What is possible?  
**Invention**

